

The AMetric Boundary as a Unique Regime-Invariant Constraint on Admissibility-Bearing Construction

Pre-fixational authority, non-transmissivity, and complete interfaces

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Abstract

The AMetric boundary constrains which distinctions and selections a determinate construction can warrant from its available data. Its basis is the necessary kernel of non-degenerate determinate objecthood: admissibility, standing, reference, and irreversibility. The pre-fixational argument excludes unlicensed representative privilege, derives lawful relabeling and equality-only dimensionlessness, and establishes the non-transmissive order of qualification. The constructed Axiom 6 excludes borrowed qualification authority without confusing binary standing with uniformity of all qualified objects. We make the dependency explicit and prove a representation-independent conservation theorem from independently warranted result incidences, without assuming agreement between competing descriptions. The mathematical boundary then has an exact form. Invariant scalar information factors uniquely through the lawful orbit quotient; an equivariant selection exists precisely when eligible stabilizer-fixed outputs can be supplied on the input orbits, with the orbitwise choice burden stated. Selection is unique exactly when every eligible fixed-point set is a singleton; a groupoid formulation extends the result to different presentation types. For an equality-only interface, finite-tuple orbits are exactly equality patterns, yielding complete scalar and selection classifications. A complete warranted observation family determines a unique maximal comparison-respecting quotient, and an arbitrary proposed interface is sufficient exactly when its fibres do not identify inputs requiring different answers. We prove exactly when the orbit and observation quotients agree and how additional queries refine the interface. These results support construction from independently licensed components and transport under faithful changes of presentation. Retained-history conservation is proved separately from a sharp endpoint theorem: endpoint standing certifies every finite legal path exactly when negative standing is absorbing. Scale fixation and self-adjoint extension selection show how additional warranted structure performs indispensable work. The boundary is universal as a necessity and conservation constraint; realization, existence, and observation coverage remain explicit mathematical burdens. No global finiteness or computability assumption is needed for the semantic theorem.

Keywords: admissibility; determinate objecthood; invariant information; equivariant selection; observation quotient; historical fixation.

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1 The problem: what can a construction claim before fixation?

A physical record, a mathematical derivation, and a reported experimental endpoint can have radically different structures. Each nevertheless makes a claim about something determinate. Its identity must survive the changes of presentation offered as descriptions of the same thing. Its warranted uses must remain attributable to the relevant bearer and conditions. A replacement of the original occurrence cannot count as preservation of its history merely because the replacement receives the same name. These are constraints on determinate construction itself.

The AMetric boundary concerns a specific consequence of that dependence: a description cannot supply additional standing-bearing discrimination merely by changing how already fixed data are represented. An output that distinguishes alternatives requires a distinction warranted by the available construction. Neither a bare name nor a numerical score supplies that warrant on its own. Yet a warranted measurement, an independently identified parameter, or a proved extremal condition can make a previously unavailable distinction. A boundary theorem should characterize this difference exactly.

The pre-metric argument supplies the upstream reason for that boundary. Before the required target-fixing discrimination has been licensed, a representative name cannot do its work merely by being named. The anti-haecceity sequence identifies the lawful indifference to representative replacement and, where equality exhausts the authoritative atom structure, derives structural dimensionlessness. The non-transmissive sequence then excludes using an interior-only metric, operator, or license as the unexplained source of its own precondition. These arguments culminate in the constructed Axiom 6 of Theorem 7.5, supported by the selection and equality classifications proved before it. The later interface, construction, and historical results develop the lawful ways to retain and supplement the warranted information.

1.1 The argument in seven commitments

The argument connects necessary governance to exact limits on discrimination, selection, and representation. Its mathematical tools are established group-action and quotient results. Their role is to expose what must be supplied independently before a construction can claim an answer. The following seven commitments give a conceptual map; their numerical order is not a claim that each theorem is proved from the preceding item alone.

1. **Actual determinacy incurs the functions.** An actual non-degenerate determinate object or incidence necessarily retains its subject, qualification, attributable use, and original incidence. These are the kernel's four functions, including for unobserved physical objects and atemporal mathematics (Proposition 2.2 and §2.2). Governance does not await adoption of an interpretive framework.
2. **An unfixed target cannot borrow its own authority.** The pre-fixational argument identifies what the complete licensed content already retains. Where equality exhausts authoritative atom structure, lawful relabeling yields equality-pattern information and structural dimensionlessness (Sections 3 and 6). The constructed Axiom 6 later combines these classifications with the fibre and grounding arguments: richer structure cannot supply a missing original distinction or license merely by presupposing it (Theorem 7.5).
3. **A particular result needs its original conditions.** The bearer and complete question must be identifiable before rival outputs are compared. The domain supplies the result relation, its domain, and its determinacy; each proposed answer requires its own warrant. Same-incidence conservation then derives agreement (Section 2). Necessity of the kernel

does not replace these substantive result proofs.

4. **Lawful transport concerns the whole incidence.** Transformations must carry the relevant inputs, outputs, and conditions according to their actual roles. Covariance of the original relation yields domain invariance and equivariance of its answer, also across different output types (Section 5). Lawfulness is established from the incidence rather than selected to protect a preferred output.
5. **Information and selection have exact boundaries.** Invariant information factors through orbits. An eligible selection must pass the stabilizer test; supplied orbitwise choices establish existence, and singleton eligible fixed-point sets establish unique selection without a choice axiom (Section 4 and theorem 5.4). Anchors and probability laws retain their own input and output types. The equality-only classification includes its finite-cardinality exceptions (Section 6).
6. **Every intended use must survive the interface.** A reduction can preserve some invariant answers while erasing a required distinction. The lossless-interface test determines sufficiency for the complete required family and declared comparison. The resulting quotient is maximal, with separate exact conditions for agreement with the orbit quotient and for refinement by added questions (Section 8).
7. **Construction must preserve the same obligations.** Partial operations must retain definedness and output classes; continuation profiles must cover the declared continuations; universal assemblies require existence and a proved universal property (Section 9). Worked realizations exhibit warranted fixing data and selection criteria (Section 10). Retained history and endpoint certification obey different, explicit conditions (Section 11).

The proofs first establish original-incidence accountability in Section 2, then develop the prefixational and symmetry classifications in Sections 3 to 6. Section 7 adds the non-transmission and grounding results and assembles Axiom 6. The interface and construction theorems then establish their own exact consequences under their displayed hypotheses; Theorem 12.1 is the final synthesis. This order keeps the classifications used to prove Axiom 6 distinct from the subsequent uses of that boundary.

1.2 Four tests for a proposed construction

The same four test names will be used throughout the paper. They identify mathematical obligations within necessary kernel governance.

The fibre test.

If two richer presentations retain the same source data, do they require the same proposed answer? Constancy on every such fibre is exactly the condition for a unique answer on the image of the forgetting map (Theorem 7.2). Total source coverage and original-incidence warrant are additional obligations.

The stabilizer test.

Is an eligible selected candidate fixed by every lawful transformation fixing its input? Theorem 4.6 gives the obstruction and the converse with supplied orbitwise choices. Theorem 5.4 distinguishes existence from exactly one selector. An irrelevant fixed point outside the eligible set does not pass the test.

The lossless-interface test.

Does equality of encoded values preserve every required answer and the declared

comparison-class distinction? Theorem 8.6 applies the fibre principle to the complete family of uses. The intended task supplies that family; a preferred success bit cannot silently replace it.

The grounding test.

Can a proposed license family be reached from independently grounded judgments by the declared sound rules? If the family has no grounded member and every rule entering it already requires one of its members, it cannot enter grounded closure (Theorem 7.4). Avoiding this obstruction alone does not establish a derivation or its semantic warrant.

The fibre and lossless-interface tests share one factorization principle: forgotten distinctions cannot be recovered as functions of the retained data. Their applications differ in the answer family and comparison being preserved. The stabilizer test concerns selection compatible with lawful transport; the grounding test concerns dependence of licenses. Randomized outputs, added queries, and transport between different presentation types are developed with their respective data and output types within this same organization. The four tests do not replace the separate existence, domain, coverage, and universal-property proofs required by particular constructions.

The complete query family may be infinite. An observer may lack an algorithm for evaluating it. The semantic conservation theorem depends on neither a finite list of encodings nor an effective enumeration of answers. The separate finite-certificate result in Section A concerns the implementations for which its additional hypotheses are established. Appendix C supplies explicit models showing why the principal mathematical hypotheses are necessary.

1.3 A running example: batteries, tests, and records

Consider two rechargeable batteries. A stated test asks whether a specified battery powers a particular lamp continuously for a stated duration under fixed conditions. The duration and apparatus belong to the physical test protocol; they are already identified and independently calibrated where calibration is required. One original trial passes and another fails. The batteries are then placed in visually identical sleeves.

A person given only the unmarked pair cannot identify the passing battery by calling one sleeve “the good one.” A leftmost choice uses an additional spatial arrangement; a serial number uses an identifier; a retest uses a new physical incidence. Each can be legitimate for an appropriate task, but its contribution must be stated and connected to the battery concerned. If the sleeves are exchanged while all information presented to the selector remains unchanged, an output that silently follows the sleeve’s former position has lost its warrant as identification of the tested battery.

Now suppose the failed battery is recharged and passes a new trial. That result licenses a statement about the new trial under its new conditions. It does not alter whether the original trial passed. Conversely, a later reading of an unchanged original record may establish that the first report was mistaken. Correcting a report need not change the original event. The original event, the record, its interpretation, and the new trial are distinct bearers of claims.

This example also prevents an indiscriminate use of “failure.” A battery that fails the specified test remains a determinate physical object. The failed trial and its record can themselves have standing as evidence of failure. What is absent is qualification for the particular successful-performance claim. A loss of standing is relative to the role and use at issue; it does not erase the objecthood of every carrier involved.

The four kernel roles are visible without imposing a new physical mechanism. Reference identifies the battery and trial. Standing concerns whether that trial qualifies for the specified

use. Admissibility keeps reliance on the trial answerable to the actual battery, conditions, and evidence. Irreversibility preserves which occurrence had which original conditions. Proposition 2.2 establishes why these roles are required; the subsequent theorems determine what their faithful representation permits.

1.4 Four forms of uniqueness

The title’s uniqueness claim has a precise object in each of four settings. First, a determinate question about one identified incidence has at most one warranted result, and exactly one when warranted existence is established. Second, exactly one eligible equivariant selector exists precisely when every input has one eligible stabilizer-fixed output, as Theorem 5.4 proves. Third, a complete family of warranted observations determines a canonical maximal presentation quotient, unique over the original presentation space. Fourth, when independently licensed data admit a universal construction, its adequate realizations are unique under the corresponding comparison maps. None of these statements identifies different questions, different source diagrams, or different physical initial data.

Regime invariance means that the constraint is preserved under faithful transport of the bearer, question, warrant, and comparison structure. It is not a claim that every regime possesses the same carrier or numerical laws. The boundary is universal because no actual determinate construction escapes the need for that work. The objects instantiated under it may differ.

The word *AMetric* has a corresponding two-level use. At the general level the constraint on authority is stated without primitive distance, order, probability, or a temporal process of authorization. At an equality-only proposal interface, it yields a complete classification of which distinctions remain before any richer authoritative structure is supplied. Logical priority to a particular fixation is not a claim that there once existed a physically structureless state. A metric can be part of a warranted physical target while supplying no independent authority to replace that target’s identity or conditions.

1.5 Relation to existing work

The constitutive argument belongs near debates about identity, individuation, and the reach of transcendental reasoning. Wiggins’s account of sortal individuation and Stroud’s challenge to inferences from requirements of thought to requirements of objects identify relevant pressures [7, 6]. Here the starting point is an actual determinate incidence. The paper does not infer the existence of the batteries or their behavior from linguistic conventions. It asks what must remain fixed for a statement about those incidences to retain its subject and warrant.

The symmetry arguments use the invariant/equivariant distinction familiar from group actions and nominal techniques [1, 2, 3]. The quotient constructions use ordinary universal properties [4]. Observation through all admitted finite continuations is closely related to behavioral equivalence and its coalgebraic treatment [5]; our partial-operation formulation includes definedness explicitly. The distinction between covariance and the structure retained under an identification is also central to philosophical analyses of gauge [8]. Here the actual comparison relation and its retained observations are specified directly. These connections locate the mathematical machinery rather than claim novelty for the elementary factorization lemmas.

The Kernel paper supplies the foundational dependency and the determinate-incidence formulation [13]. The necessary commitments are restated and argued here, and all boundary and interface proofs are given below. Downstream corpus papers motivate the examples and

the order of construction, but their terminal uniqueness conclusions are not premises of this paper.¹

The local pre-fixational treatment in *The Structure of Admissibility* makes equality-only authoritative structure and backward non-transmission explicit.²

1.6 What would defeat the argument?

The constitutive argument is challenged by showing that actual determinate objecthood does not entail one of the functional commitments identified in Proposition 2.2, or that the reconstruction of that commitment misidentifies the work done by a proposed construction. Such a challenge must engage the entailment from objecthood, not demand that a role retain its work while omitting that same work. If an alternative mechanism does perform the required function, renaming or distributing it supplies an implementation of the role. The mathematical claims have sharper counterexample conditions: a purportedly lawful transformation that changes an independently warranted answer; a claimed complete interface that identifies inputs requiring different answers; an equivariant selector with no eligible stabilizer-fixed output; or a claimed universal construction whose comparison maps do not have its stated universal property. These tests refer to original data and functions, so an inconvenient rival cannot be dismissed simply by adding a new clause to “same scope.”

2 Determinate incidence and necessary kernel governance

2.1 The target before its representation

A construction is *non-degenerate and determinate* in the relevant sense when it sustains an actual identity-bearing object or incidence whose conditions ground at least one correct identification or qualified use, whether or not anyone performs that identification or use. This excludes a mere inscription claiming those features. It does not require every proposed operation to succeed or every question to have an answer. The domain’s substantive conditions determine what succeeds; the kernel identifies the work that any such determination necessarily performs.

Definition 2.1 (Complete locus and original incidence). A complete locus is a bearer b together with a question q specifying its relevant occurrence, conditions, and intended use. A result incidence is a relation

$$J(b, q, v), \quad v \in V_q,$$

fixed by the original domain rules or actual conditions of the incidence. Its specification precedes the comparison of proposed answers. It is *determinate at* (b, q) when

$$J(b, q, v) \wedge J(b, q, w) \implies v = w. \quad (1)$$

The bearer and question must be identifiable without inserting the proposed result as a device for forcing (1).

¹The source developments include *Constraint Architectures for Coherent Evaluability*, *Exact Certificate-Fibre Exhaustion*, *The Structure of Coherent Construction*, and *Metric-Relative Exactness*. They are contextual manuscripts [14, 15, 16, 17]. No downstream assertion that assumes the AMetric boundary is used to prove that boundary here.

²See [18, Definition 4.1, Theorem 4.25, and Lemma 4.30]. The present paper supplies the arguments and the broader factorization and grounding proofs directly. The source is evidence of the boundary’s intended dependency, not a substitute for a proof. Downstream applications in *Metricity as Licensed Description* and *Unus Solus Possibilis Est* use the boundary upstream [19, 20]; their conclusions are not used here to prove it.

There can be many possible results before a particular outcome is fixed. Then a question about the possible-outcome set may be determinate while a question selecting one future outcome is not. Likewise, a specified battery and a specified circuit do not alone determine every proposed numerical question about their behavior. Equation (1) requires a proof or domain warrant for the particular result relation. The argument below never infers that equation from numerical identity alone.

Proposition 2.2 (Constitutive kernel dependence). *Every non-degenerate determinate object or incidence necessarily performs the work of reference, standing, admissibility, and irreversible fixation. A physical domain containing such objects, states, carriers, records, or endpoints therefore instantiates these roles without a further interpretive authorization.*

Proof. A claim to a determinate bearer distinguishes that bearer from a substitution. Reidentification under a change of presentation must preserve what the claim concerns. This supplies the reference work *Ref*.

An actual incidence under stated conditions distinguishes the relations that hold there from relations that would hold only of another bearer or under other conditions. Its fitness for a specified identity-bearing use is answerable to that distinction. This is the standing work *St*. A name or observer's endorsement cannot supply the missing relation.

Using the incidence in a further identification, inference, comparison, or construction must retain that qualification and its bearer. Otherwise what is carried forward can be a different incidence while being offered as the same one. The joint constraint on warranted reliance is the admissibility work *Adm*.

Finally, the original occurrence and its original conditions remain the subject of an historical claim. Replacing either changes that claim's subject or conditions, even if a later performance succeeds. Retaining this distinction is the irreversibility work *Irr*. In an atemporal construction, the relevant fixation is an identified position or instantiation; no physical time evolution is required.

Removing any work described above loses the corresponding identity-bearing commitment. Implementing it under another name retains the work. Non-degeneracy ensures that an actual bearer grounds the qualification for use, rather than a vacuous set of purported conditions. The qualification need not have been inspected, known, or exercised. Thus the four functions are incurred by determinate construction itself. $\square$

This is a constitutive argument about actual identity-bearing incidence; the existence of an uninterpreted element alone does not establish it. Section 2.2 defends that inference and distinguishes actual qualification from its recognition.

The functions can be combined or distributed; their necessity does not posit four distinct controllers or an additional physical interaction. The table identifies their work, and Section 2.2.7 supplies the reconstruction argument and its precise sense of minimality.

Role	Required work	Loss in the battery example
Reference	Preserve the bearer and complete query	A different battery or trial answers for the original.
Standing	Retain qualification for the specified use	A positive label substitutes for actual test performance.
Admissibility	Keep further reliance tied to that qualification	Evidence about one trial is used to authorize an unsupported claim about another.
Irreversibility	Retain the original occurrence and conditions	Recharging and retesting are reported as changing the original failure.

These losses can overlap. The separation is functional, not a demand that each role occupy a different piece of machinery. The failed battery remains kernel-governed as a determinate object, and a sound report of failure can have standing in its evidential role.

2.2 Why the four roles are constitutively necessary

Two steps must be distinguished: why determinate objecthood incurs the commitments, and how those commitments reconstruct the four roles. Defining an adequate implementation to contain the roles cannot establish the first step. The argument below supplies that defense directly, drawing on the Kernel's objecthood and reconstruction argument [13, Proposition 2.4 and Theorem 3.1].

2.2.1 From a determinate subject to reference and qualification

Begin with an actual object or incidence, rather than a description that merely announces one. To identify a particular battery trial is to concern that trial rather than another trial that happens to receive the same name. This contrast is present in the identification itself. If replacing the battery, load, or occurrence makes no difference to what the identification concerns, the claimed specificity has disappeared. Conversely, an inscription can change without changing the identified trial. Reference is the work of retaining the subject through that distinction between a change of presentation and a substitution. An observer need not possess a complete method for recognizing every substitution for the distinction to hold.

Qualification concerns a second question. The identified arrangement either supplies the relation claimed of it under the specified conditions, or the claim is not supported by that arrangement. A trial's having carried the specified load cannot be supplied by a different trial's having carried it. This is not yet a conclusion that every question has a unique result. Qualification can concern membership in an outcome set, satisfaction of a relation, or one selected result when the domain establishes such a result. What is required is that the claim answer to the identified subject and its actual conditions, rather than to an endorsement detached from them.

Standing names that qualification for a specified role. The successful performance qualifies for the success claim; a faithfully identified failed performance can qualify for the claim that the test failed. Those are different uses of determinate incidences. Neither the trial's objecthood nor the standing of a true failure report is removed by the battery's failure to power the lamp. This distinction is indispensable: otherwise a theory of standing would conflate an unsuccessful practical purpose with the absence of anything determinate to explain, diagnose, or record.

Why does objecthood incur qualification at all? Because identification is already answerable to what is identified. Even where no practical test is selected, an alleged identification concerns this bearer only if the actual identity conditions support that attribution. If no identification or identity-bearing use could be correct of the purported object, there is no non-degenerate determination of it in the sense under investigation. This is the substantive constitutive claim. It does not say that an object exists only when someone succeeds in describing it. It says that an actual determinate object supplies conditions to which a correct identification would answer. The qualification precedes its recognition.

2.2.2 Actual qualification, evidence, and physical existence

Three matters should consequently remain distinct: what obtains, what evidence is available, and what an investigator is justified in reporting. An unread record can be correctly related to an earlier trial. An inaccurate report can concern a determinate record. A justified provisional

estimate can subsequently prove inaccurate. The first is an incidence between record and occurrence; the second is an assessment of a report; the third concerns an evidential procedure and its limits. Calling all three “standing” without their complete uses would conceal the very distinctions the kernel requires. Each may instead be assessed at its own bearer and question.

An incomplete report can erase a real distinction without making the original incidences interchangeable. Sections 3.3 and 8.6 distinguish such information loss from a symmetry of the complete input and give the explicit battery model.

The same distinction answers the unobserved-object objection. An unobserved battery trial has its actual arrangement and relations whether or not a reader is available. Human purposes can select the question whether its performance meets a specified protocol. They do not cause that historical performance to have met it. A change of protocol can yield another correct assessment, but cannot change which relation held under the first protocol. The object’s necessary governance is therefore not postponed until an observer adopts an assessment vocabulary. What observation adds is access to a qualification, or a new physical incidence with its own qualification.

This argument avoids an inference from norms of description to the existence of physical objects. Such an inference would face the familiar objection that requirements of thought need not establish corresponding features of an independently existing world [6]. Here physical existence and the substantive relation are supplied by the domain. The constitutive argument examines what their determinacy entails. It does not derive the battery, a law of electrochemistry, or a measurement result from rules about sentences. The physical relation explains why a proposed statement can be correct of that battery; correctness is not invoked to create the relation.

Thus “warrant” has a semantic ground and, when a claim to knowledge is made, an additional evidential burden. A proof of the conservation theorem must establish the incidences it compares. An observer who lacks their evidence cannot invoke their truth merely by writing a warrant field. Equally, missing evidence does not change the relation’s actual qualification. Keeping these directions separate preserves both physical independence from our assessments and the accountability of our assessments to physics.

2.2.3 Why admissibility is more than a favorable label

Standing and admissibility can be implemented together, but they answer different questions. Standing concerns whether this incidence qualifies for this use. Admissibility concerns whether the proposed reliance on it retains the conditions under which it qualifies. A correct original record can be misused. For example, a record establishes that battery b passed under load L for duration d . Citing it as evidence that battery b' passed, or that b met a different load and duration, requires further support. The record need not lose its standing as the original record for that particular inference to fail.

The need for this distinction follows from composition. A premise carries information about specified relata under specified conditions. A conclusion that relies on it must either retain those conditions or justify the transition to different ones. If the connection disappears, the conclusion is no longer supported by that premise in the claimed way. A symbol saying “passed” does not restore it. Admissibility is the requirement that the constructional connection be warranted. It does not require every transformation to be an identity transformation: an operation can construct a new object, provided its application and the new object’s qualification are established as such.

This also distinguishes admissibility from an external controller. A proof rule, a typed operation, a physical coupling, or an identification map can perform the relevant work without

a separate gate device. The constraint concerns whether the connection actually holds, not whether a mechanism consults a permission label. Physical interaction itself can occur when an investigator's proposed inference about it is mistaken. In that case the interaction remains a determinate physical incidence, while the inference lacks its claimed standing. The kernel does not forbid the event in order to criticize its description.

The distinction is functional rather than a claim of logical independence. One rich relation might record both qualification and every licensed way of using it. Such a representation performs both roles. The distinction remains useful because its two failure questions differ: did the original incidence qualify, and did this reliance preserve or legitimately transform that qualification? An answer to the first does not settle the second without the connecting argument.

2.2.4 Fixation without a universal arrow of time

Reference identifies the subject currently under discussion. Irreversible fixation retains which subject and conditions an already specified incidence concerned. Their work overlaps, especially when occurrence information is included in a rich identity type, but the second question cannot be replaced by a current-state description. Two trials can finish in the same observed arrangement while having different loads or interruptions. A statement about the first trial must retain its original conditions even if the current arrangement is indistinguishable from the second trial's endpoint. Otherwise equality of endpoints would silently become equality of histories.

This is why reversal does not defeat fixation. Suppose a physical evolution is invertible and later returns a system to its earlier state. That return can be a genuine physical achievement. It does not make the later return the earlier occurrence, or make an intervening unsuccessful trial never have occurred. The distinction concerns the content of the original historical claim. It does not assert that a physical record cannot be erased, that all dynamics are dissipative, or that entropy must increase. Destroying evidence can remove access to a historical answer without supplying a different answer to the unchanged historical question.

An atemporal mathematical object requires no history of physical creation. Its relevant fixation can instead be the argument of a function, an instantiation of a rule, or a position in a derivation. Replacing that argument while attributing the result to the unchanged application would change what was evaluated. One may legitimately form the new application; one may not attribute its warrant to the old application without the required relation. The original and new mathematical objects can coexist without any temporal order. Fixation here retains an identified incidence, not a cosmological or thermodynamic direction.

The minimal historical conclusion is therefore non-substitution at a fixed locus, as in the Kernel's historical-fixation lemma [13, Lemma 3.3]. The stronger claim that a negative state can never reach a positive state is different. Its exact additional premise is proved in Theorem 11.2. Retaining a failed original trial does not prohibit constructing a successful new one. This distinction is necessary to defend irreversibility without turning ordinary repair into a counterexample to determinate objecthood.

2.2.5 A determinate mathematical object with no physical test

Consider the integer 2 in the usual integer structure, and the application of the successor operation to it. The structure and operation are fixed before a numeral is evaluated. The question asks for the successor of that integer; the arithmetic rules establish the answer 3. A second notation for the same input is acceptable only through its interpretation in the same structure. Another occurrence of the character "2" in a different coding system need

not denote that integer. Reference concerns the actual value and operation, rather than the typographic shape.

Standing concerns the equation's qualification under those arithmetic rules. Admissibility concerns, for example, a subsequent substitution of equal integers into that operation or the use of the equation in an argument. A valid substitution preserves the established relation; an unsupported substitution of a different input is a different application. Fixation retains which application the original calculation evaluated. Changing the input to 3 and obtaining 4 does not change the result of the successor application to 2. None of this requires an experiment, a physical clock, a permanent inscription, or an observer actually performing the calculation.

The example does not derive arithmetic from the kernel. Arithmetic supplies the structure, the operation, and the proof of the result. The kernel identifies what that proof already accomplishes as a determinate construction. Nor does the example presuppose that every string denotes an integer or every mathematical question is decidable. A malformed expression can be a determinate string while lacking standing as the alleged integer expression. A difficult true statement can retain its truth when no proof is available to a particular investigator. The bearer and the claimed use must remain distinct in each case.

The objection that fixation merely redescribes identity is partly correct about implementation: a sufficiently rich identity specification can encode the whole incidence. It does not follow that the fixation work has vanished. The specification now preserves both the input's identity and which application concerned it. The four-role claim concerns these requirements on determinate use, not four irreducible ontological parts of the integer.

2.2.6 Individuation, revised evidence, and legitimate disagreement

The requirement to fix a bearer is not a method for deciding disputed individuation. A record may not disclose whether two observations concern the same battery, two stages of one trial, or different trials. Identity conditions themselves can require physical investigation or a mathematical proof. Conditions of individuation rather than mere persistence of a name are therefore central; this is an appropriate point of comparison with work on sortal identity [7]. The present theorem does not settle every such dispute from the word "reference." It requires a claimed comparison to expose the identification on which it relies.

Revised evidence can then have several effects. It can correct the assigned bearer, disclose an omitted condition, or establish that the old bearer and conditions were already as claimed. Only the last is straightforwardly a better-supported answer to the unchanged complete question. The first two may reveal that a formerly advertised same-locus comparison was mistaken. These possibilities must be settled by evidence concerning the original incidence. Disagreement between outputs alone proves neither that the bearer changed nor that one answer lacks warrant; those conclusions need the established identity and result conditions.

Consequently the account permits genuine criticism. A critic can challenge whether the proposed reference conditions identify any determinate object, whether qualification follows from the actual relation, whether a use retains its supporting conditions, or whether the historical attribution has been reconstructed correctly. The critic need not accept the conclusion by adopting a definition of adequacy. The necessity argument stands or falls at those constitutive inferences. Once those inferences are justified for determinate objecthood, however, terminology is no further permission condition: a physical or mathematical realization performs the work regardless of what its theory calls it.

2.2.7 Reconstruction and the precise force of minimality

Reconstruction examines the actual dependencies that identify the bearer, ground its qualification, connect a use to those conditions, and retain the original incidence. The work may be distributed across components or compressed into one relation. Naming a field after a kernel role does not show that it performs the work; deleting the name does not show that the work has disappeared.

Suppose a realization still supplies the complete determinate job after a claimed functional deletion. If another mechanism performs the required work, the function remains. If none does, the corresponding identification, qualification, reliance, or original attribution is lost. This minimality argument depends on the preceding necessity argument; it is not a second proof of it [13, Theorem 3.6].

Functional minimality does not establish logical independence of four axioms. A single axiom, program, or physical structure may perform several roles. An independence theorem would require explicitly formulated axioms and independence models. The present claim concerns indispensable work, rather than a fixed number of fields or primitive symbols.

These functions do not select a unique physical law or admission policy [13, Proposition 4.4]. Every actual non-degenerate determinate incidence incurs the four roles, while its existence, result determinacy, and individual answer warrants require their own domain arguments. The following conservation theorem derives agreement from those warrants; later orbit and quotient theorems cannot supply the missing domain facts or establish the constitutive argument backwards. Section 12.1 traces these dependencies through the full boundary theorem.

2.3 Agreement derived from separate warrants

Definition 2.3 (Accountable presentation use). A presentation use supplies an identification of the original bearer b , the complete question q , and an output $v \in V_q$. Its claimed standing as an answer requires a warrant for $J(b, q, v)$ from the original incidence conditions. Identification of its bearer and query is independent of agreement with another presentation's output.

Theorem 2.4 (Same-incidence conservation). *At a determinate locus (b, q) , any two accountable standing-bearing answers have the same result. This holds for every representation type and every pair of actual presentation uses identifying that locus.*

Proof. Let the outputs be v, w . Their individual warrants establish $J(b, q, v)$ and $J(b, q, w)$. Equation (1) yields $v = w$. Where different names are used for equal query indices, first transport both outputs to the same result type. No equality of outputs was used to establish the bearer, query, or either warrant. $\square$

The proof depends on three independently established premises: the original relation is fixed, its determinacy is justified, and each proposed answer has an incidence warrant. One may prove a parser sound, identify an actual trial, or prove a domain theorem furnishing that connection. Defining a certificate field named “sound” does not discharge it. Conversely, a rival that supplies a different warranted answer at exactly the same determinate locus defeats one of the stated premises; it cannot be dismissed by renaming the locus after inspecting the answer.

Corollary 2.5 (No discordant co-authority). *At a determinate locus (b, q) , if $J(b, q, v)$ is established and $w \neq v$, then w has no standing as an answer to that same complete question about that bearer. Another encoding, formal language, or physical reader cannot supply an independent co-authority for w there.*

Proof. An incidence warrant for w would contradict Theorem 2.4. $\square$

This is a universal exclusion of discordant authority, not a restriction to encodings listed by a grammar. A newly found proof can establish our access to an already true result. A new experiment can fix a new result. Neither is excluded. What cannot occur is the acquisition of a different warranted answer to the unchanged determinate question merely by changing its representation.

Theorem 2.6 (Complete-profile conservation). *Let $\mathcal{Q}_b$ be an independently specified family of complete questions about b , each with determinate incidence. Suppose presentations r and r' supply answers $F_r(q), F_{r'}(q)$ with original incidence warrants for every $q \in \mathcal{Q}_b$. Then*

$$F_r = F_{r'} \quad \text{in} \quad \prod_{q \in \mathcal{Q}_b} V_q.$$

Proof. For any fixed q , Theorem 2.4 gives $F_r(q) = F_{r'}(q)$. Equality follows by function extensionality. The proof imposes no finiteness or effectivity condition on $\mathcal{Q}_b$. $\square$

Coverage of $\mathcal{Q}_b$ is part of an application claim. If a task observes rejection reasons, continuation options, or original conditions, those distinctions must appear as coordinates. If only a fragment has been warranted, the theorem applies to that fragment. The full profile is not a single bit merely because one admission question is bivalent.

2.4 Bivalence, definedness, and evidence

For a complete qualification proposition $A(b, q)$, classical semantics supplies its characteristic value in $\{0, 1\}$. It is true or false under those conditions, not partially true because a score is near a threshold. This is semantic bivalence of the qualification question. A score may nevertheless be a warranted input to the proposition, as the independently fixed test duration is in the battery protocol. Neither score variation nor differing reasons for failure introduce a third truth value.

Exactly two *realized* standing classes require both a positive and a negative instance. For one fixed question whose answer is positive, its answer space need not contain an actual negative instance. We use no surjectivity onto $\{0, 1\}$ without that additional witness. Classical bivalence also does not supply a decision algorithm or make an unfinished computation a warranted negative.

Definition 2.7 (Complete outcome of a partial incidence). Let $J_0(b, q, v)$ be a determinate base result relation. Define outcomes in $V_q \sqcup \{\perp\}$ by

$$\begin{aligned} \hat{J}(b, q, v) &\iff J_0(b, q, v), & v \in V_q, \\ \hat{J}(b, q, \perp) &\iff \neg \exists v \in V_q J_0(b, q, v). \end{aligned}$$

The tag $\perp$ here reports semantic non-incidence, not an elapsed time limit or lack of a known proof.

Proposition 2.8 (Conservation includes definedness). *The relation $\hat{J}$ is determinate. Ac-countable complete-outcome presentations agree both on definedness and on the value when present.*

Proof. Two present results agree by determinacy of J_0 . A present result and $\perp$ cannot coexist, since the latter denies the former's incidence. Two absent outcomes have the same tag. Apply Theorem 2.4. $\square$

The carrier of an unsuccessful search is still a determinate object. Its accurate report may be “not yet established.” That report concerns search status, a different query from semantic non-incidence. This distinction is indispensable when an interface is claimed to preserve failures. A sound search returning only positive results is not yet a complete-outcome evaluator.

2.5 The boundary and its implementations

The AMetric constraint is expressed by relations, equivalences, and universal properties. It does not require an additional event in which a metaphysical gate acts on the world. A physical checker or measuring device is an ordinary determinate construction and itself incurs the kernel. Distinguishing the constraint from its implementation does not remove governance from the implementation; it prevents the abstract relation from being mistaken for an extra physical cause.

The next sections identify the exact information and selection structure compatible with these commitments. They do not assume that a candidate transform preserves the answer merely because it has been called a redescription. Its effect on the original bearer, query, and warrant must be established first.

3 Pre-fixational authority and the pre-metric constraint

The necessity of the kernel leaves a further question to be answered. What can supply the first target-identifying authority of a proposed construction? A name can identify a position in a record, an algorithm can choose a representative, and a metric can compare already specified relata. Their existence does not yet establish the relation by which their output concerns the proposed target. The pre-fixational argument examines that missing work. It does not begin by denying that names, algorithms, or metric-bearing physical systems exist. Each is itself a determinate construction and necessarily incurs the kernel.

The argument has two levels. First, representative identity cannot supply an additional warrant for an unchanged original incidence. This anti-haecceity constraint applies wherever the relevant content and identification have been independently fixed. Second, when equality exhausts the authoritative structure of the proposed atom interface, its complete relabeling symmetry gives an exact structural restriction. The second level requires its stated content hypothesis; the absence of a named individual alone does not establish it. Keeping the levels separate preserves the constitutive argument while exposing the full mathematical burden of its strongest specialization.

3.1 Fixation is relative to a target role

Definition 3.1 (Target fixation and pre-fixational proposal). For a specified use, target fixation is the establishment of the relation that makes a proposed bearer determinately identifiable as the target of that use, under its complete question and conditions. A proposal is pre-fixational with respect to that role when this target-identifying relation is not yet among the grounds available to the construction in that role. Its symbols or physical carriers may already be fully determinate. Their identity as carriers is not, by itself, their qualification as the proposed target.

The qualification “with respect to that role” is essential. A written variable is a determinate mark before it has been interpreted as the argument of a particular function. A labelled

sample tube is a determinate physical object before a proposed identification with an original specimen has been established. A set-theoretic atom is distinguishable in the metatheory before its identity has been licensed to settle a question in an equality-only interface. In each case the carrier's reality and the target-identifying relation are different incidences. The kernel governs both; the first incidence cannot silently perform the second's work.

This distinction is not the claim that targets await human recognition in order to exist. An actual relation fixing a trial's battery remains actual when no observer knows it. If that relation belongs to the complete original conditions, it must be retained even when a report omits it. The report is then incomplete; ignorance has not made the physical target pre-fixational. The present question is instead whether a construction offered as the source of an identification has the relation needed to support that identification. An unread identification record and an arbitrary label without any identification relation are different cases.

Nor must fixation occur as an earlier event in physical time. An interpretation, an equation's domain, and its result can be specified together in one mathematical construction. What matters is the dependency: the equation cannot support a claim about its intended argument while omitting the interpretation that makes it that argument. Logical priority here concerns what an attribution requires, not a period during which an actual physical carrier exists without governance. The word "first" marks the first licensed target-identifying role in the argument; it does not posit a first event in the universe.

3.2 Anti-haecceity at an independently identified incidence

Definition 3.2 (Representative privilege and licensed discrimination). A proposed ground of authority is haecceitic in the relevant sense when it treats the bare identity of a representative—this inscription, token, index, or encoding—as an additional warrant while all independently licensed content relevant to the original incidence is unchanged. A licensed discriminator instead has a warranted relation to the bearer, question, or conditions that actually distinguishes the cases at issue. That relation may identify an individual, distinguish a configuration, or preserve a relational feature without selecting any individual.

Thus an index is not intrinsically illicit. A registry can warrant which specimen an index denotes. The warrant comes from the registry's actual identification relation, together with the conditions of its use; the printed index does not generate that relation. Conversely, a change of label that retains the very same identification supplies no new authority to answer the unchanged question differently. Anti-haecceity concerns that unsupported promotion of representative identity. It does not forbid ordinary reference or require all distinct objects to have the same properties.

Proposition 3.3 (Equal-content, fixed-locus anti-haecceity). *Suppose two presentation uses have been independently identified with the same bearer b and complete question q , with all relevant original conditions retained. Let $J(b, q, v)$ be determinate. If their outputs v, w each have an original-incidence warrant, then $v = w$. Adding or changing a representative name alone cannot create a second warranted result at that locus.*

Proof. The separate warrants establish $J(b, q, v)$ and $J(b, q, w)$. Determinacy yields equality, as in Theorem 2.4. The identification of the bearer and question and the retention of their conditions precede this comparison; output agreement is not one of their premises. A further name either preserves those premises, in which case the proof is unchanged, or has a separately established relation changing the relevant content. The latter is an additional datum whose role requires its own warrant, not authority created by bare naming. $\square$

The proposition identifies the constitutive reason for label neutrality. A warrant answers to the relation that holds of its subject. If representative identity could alter the warranted answer while that relation and complete question remained unchanged, the proposed authority would no longer answer to the identified incidence. This contradicts the role in which it is offered. The exclusion is therefore stronger than a stylistic preference for symmetric notation, but more precise than an assertion that all descriptions or all targets must agree.

There are substantive premises to verify. “The same content” must include every independently authoritative condition relevant to the claimed use, including relational structure, distinguished parameters, and the interpretation of the question. A comparison may not erase an inconvenient original condition and then describe its erasure as neutrality. Equally, disagreement alone does not prove that the conditions changed. The original identification and incidence evidence must establish which claim failed. This prevents the argument from protecting a favored answer by defining every rival as a different target after the fact.

3.3 From unchanged content to the correct relabeling group

A pre-fixational comparison concerns the authoritative content that the construction actually retains. Suppose that content contains a relation R on a carrier P . A passive renaming carries both the elements and the relation to a new presentation. An active permutation of P with R held fixed is a symmetry only if it preserves R . Confusing these operations would turn ordinary covariance under renaming into the false claim that every permutation preserves every structured object.

An unmarked four-cycle gives a decisive example. No vertex is distinguished, and the automorphism group acts transitively on vertices. Nevertheless adjacency distinguishes pairs, and graph distance can be constructed from the retained adjacency relation. A permutation sending adjacent vertices to opposite vertices changes authoritative content. The absence of a preferred individual neither removes that relation nor establishes full permutation symmetry. A licensed relation can distinguish configurations before it supplies an individually selected target.

The exact additional premise can now be stated.

Definition 3.4 (Equality-only authoritative atom interface). Let P be a set of candidate atoms used in a specified proposal role. The interface is equality-only when equality exhausts its authoritative atom structure: no name, relation, coordinate, distinguished subset, order, parameter, or background association supplies a further atom-specific or configuration-specific distinction for that role. For a tuple input, its coordinate positions and its equality pattern are retained. This is an assertion about the complete content of the proposed interface, not a claim that its actual carrier has no other properties.

An application must justify this assertion from the original use. It is appropriate when candidate tokens supply only positions for possible relata and no further relation has yet been licensed to distinguish them in that role. It is inappropriate when adjacency, an actual trial assignment, a marked endpoint, or any other relevant structure is already retained. No observer may obtain the premise merely by failing to discover such structure. The construction’s claimed authoritative content, and the incidence it purports to support, determine the audit.

Proposition 3.5 (Relabeling indifference from retained content). *At an equality-only authoritative atom interface, every bijection of P preserves all retained atom structure. Suppose the corresponding change of presentation identifies the same complete incidence and transports its result type as specified by the original interpretation. Separately warranted results then agree after that transport. In particular, every independently warranted scalar admissibility-relevant predicate on a finite tuple input is invariant under the componentwise action of $\text{Sym}(P)$.*

Proof. Every bijection preserves and reflects equality. It therefore preserves precisely the authoritative structure supplied in Definition 3.4, including equality among tuple positions. By the stated original interpretation, this retained-content comparison identifies the same complete incidence. Proposition 3.3 gives agreement of its warranted results after result transport. For scalar results the transport is the identity, so the predicate’s truth value is unchanged. All bijections qualify because no further authoritative relation restricts them. $\square$

Invariance is obtained from unchanged licensed content and accountability to the original incidence. It is not introduced by defining every desired answer to be invariant. The content and interpretation hypotheses do genuine work: they can fail in the cycle example or in a report omitting which battery actually passed. With additional retained structure, the same argument uses its actual automorphism group and the appropriate output transport, rather than $\text{Sym}(P)$. The general relation-based transport theorem is Theorem 5.2.

This is the faithful form of the claim that a non-invariant pre-fixational predicate presupposes an individuator. Under a complete equality-only account, two tuples of the same equality pattern differ only in representative assignment. A scalar rule separating them uses a distinction the stated account does not supply. To warrant that distinction, one must establish an additional relevant relation or revise the allegation that the original content was complete. Neither merely writing the rule nor renaming it a foundational constraint supplies the missing relation.

3.4 Canonicalization and provisionality do not supply their own warrant

Canonicalization can perform two different tasks. A canonical class code records which equivalence class an input belongs to. A canonical representative selects an element of that class. The first does not automatically accomplish the second. The orbit projection in Theorem 4.2 is an invariant class code; its existence is entirely compatible with failure of every eligible equivariant selector in Theorem 4.6.

For example, an unmarked pair can support the answer “two distinct candidates.” That answer does not identify either candidate as the original target. If exchanging the candidates preserves the complete selector input, it exchanges its eligible individual answers. Neither answer is fixed by the input symmetry, so a lawful individual selector is unavailable from those data. Replacing the pair by a code for its class supplies no missing target association. A record identifying one member, by contrast, changes the available data through a relation that can be independently warranted.

A procedure selecting the lexically first label can also be perfectly well defined. Its label order is additional input or a representational convention. If changing that order changes which physical candidate is reported, the procedure has not established that each choice answers the unchanged original target question. If its output is merely a normal-form encoding and carries no such target-selection claim, no contradiction follows. The distinction turns on the proposed use and output type, not on whether the procedure is called canonical, natural, or computationally efficient.

Proposition 3.6 (Canonicalization does not create a missing target association). *A code constant on the available-data classes cannot recover an original answer that differs within one such class. An eligible individual selector cannot be equivariant at an input whose stabilizer fixes no eligible output. Supplying a convention or anchor can change either test only through the additional information and warranted associations that it actually contributes.*

Proof. If two inputs have the same code, every function of that code gives them the same answer; unequal required answers therefore cannot both be recovered. For selection, equivariance forces

the selected output to be fixed by every symmetry fixing the input, giving the second assertion. Additional data enlarge the input. Their effect must be assessed there; relabelling a missing association as a convention does not make it present in the old input. $\square$

Provisionality concerns a different issue. A proposed distinction may later be revised, but its present use still has a definite role. If it is offered now as identifying the target or licensing an exact answer, its revision policy does not establish the required current relation. A provisional selector is a selector for that use; a defeasible ranking still ranks the candidates on which its current decision depends.

Proposition 3.7 (Revision policy does not discharge a present incidence obligation). *For an unchanged complete question, every version of a presentation offered as a warranted answer must satisfy the original incidence relation. Declaring a version provisional, approximate, defeasible, or revisable does not replace that obligation. Versions separately warranted at the same determinate locus agree there.*

Proof. The revision declaration specifies how a later version may be produced or withdrawn. It does not establish the present bearer–question–result incidence. If that incidence is supplied for two versions, Proposition 3.3 applies; if it is missing, a future revision policy supplies no instance of it. $\square$

This leaves estimates, confidence assessments, and revisable hypotheses available for the uses their evidence actually supports. A warranted statement about an error bound or present uncertainty is a different complete question from a claim that an exact original answer has already been identified. The point is not that provisional reasoning lacks all standing. Its qualified role must remain visible when its result is used.

3.5 Dimensionlessness as an exact structural restriction

Here “dimensionless” describes the absence of licensed atom-specific geometry or grading beyond equality/disequality in the authoritative atom interface. It does not mean merely that a numerical ratio has no units. A dimensionless ratio can carry extensive structure inherited from identified quantities, operations, normalization, and interpretation. Nor does dimensionlessness mean that no number, relation, or function can be written. Its exact content is what the equality-only interface can support without an additional discriminator.

Proposition 3.8 (Equality-only structural dimensionlessness). *Let P be an equality-only authoritative atom set and let tuple positions be fixed. For every finite arity:*

- (i) *scalar invariant information is exactly information about the realized equality patterns;*
- (ii) *if P is nonempty, every invariant scalar function on individual atoms is constant;*
- (iii) *if $|P| \geq 2$, no invariant asymmetric relation ranks distinct atoms, and every real-valued invariant metric on P has one positive value on all unequal atom pairs;*
- (iv) *an equivariant atom output from a tuple is a used atom selected by its equality-pattern block, or the unique unused atom when exactly one remains. These choices can vary with the pattern;*
- (v) *if $|P| \geq 2$, no equivariant selector chooses a member of every nonempty subset of P , and no atom can be selected equivariantly without input.*

Proof. Two finite tuples have the same equality pattern exactly when the correspondence between their used atoms is bijective. Extend that correspondence to a permutation of their finite union, and act identically outside it. Hence the equality patterns are exactly the orbits, proving (i). There is one pattern in arity one, giving (ii). A transposition reverses any purported asymmetric comparison between distinct atoms; transitivity on unequal pairs makes every invariant metric constant there, with positivity supplied by the metric axioms. This proves (iii).

The stabilizer of a tuple fixes each used atom and permutes the unused atoms arbitrarily. Its fixed atoms are therefore the used atoms together with the unique unused atom, if one exists. Transport among tuples of the same pattern gives precisely (iv), as developed in Theorem 6.7. For (v), the setwise stabilizer of a pair exchanges its two eligible members; without input the full symmetric group moves every candidate. The fixed-point criterion forbids the claimed selections. The full equality-pattern proof appears in Theorem 6.3. $\square$

The exceptional cases are part of the result. On an empty atom set, positive-arity tuple domains are empty; there is a unique empty atom-output function. The empty tuple still exists, but cannot return an atom from an empty set. On a singleton set, input-free atom selection is unique and the only metric is zero. For larger finite sets, a tuple using all but one atom determines the remaining atom without naming it. Coordinate projections are equivariant because coordinate positions are retained. None of these cases supplies an unlicensed identity distinction.

Finite relational information also survives. Three coordinates can exhibit several equality patterns, and the number of distinct entries is an invariant scalar statistic. Consequently “only equality structure” does not mean “only two tuple classes” or prohibit every relational grading. Likewise, the discrete metric with common off-diagonal value $c > 0$ exists; equality-only data supply no preference among such scales. On a nonempty finite carrier the uniform probability law is invariant, but it does not select a particular atom or certify a particular original answer. These distinctions prevent structural dimensionlessness from becoming a false ban on all mathematical structure.

Bivalence has a separate role. A fixed qualification proposition has two possible truth values, but a scalar predicate on one transitive atom orbit is constant. Both polarities need not be realized there. Different equality patterns, questions, conditions, or independently identified bearers can carry distinct information while every qualification remains bivalent. Neither the equality-only classification nor the absence of an individual selector proves that all positive constructions have one complete profile.

3.6 Logical construction cannot manufacture a missing discriminator

The restriction also survives an explicitly specified language of construction. This is a positive closure theorem for formulas, rather than an assertion that every possible constraint must use a metric, a temporal process, or a probability. Its scope is the given primitives and grammar.

Lemma 3.9 (Invariance under first-order construction). *Let a group G act on a nonempty set P . Interpret a first-order language on P so that every primitive relation is invariant, every primitive function is equivariant, and every constant is fixed by G . For a variable assignment a , write $(ga)(z) = g(a(z))$, and let $\text{val}_a(t)$ denote the interpretation of a term under a . Then every term t and formula φ satisfy*

$$\text{val}_{ga}(t) = g \text{val}_a(t), \quad \varphi[ga] \iff \varphi[a].$$

In particular, a formula whose free variables lie in the finite tuple $(x_1, \dots, x_n)$ defines a G -invariant relation on P^n .

Proof. Proceed by structural induction. A variable has the term property by definition of ga ; a constant has it because its interpretation is fixed. For a compound term, apply the induction hypotheses to its arguments and then equivariance of its primitive function. An atomic relation is preserved by the term property and invariance of that relation. Equality is preserved and reflected because g is bijective. The Boolean connectives preserve equivalence of truth values.

For the existential step, suppose $\exists z \psi$ holds under a , witnessed by $u \in P$. The induction hypothesis applied to the modified assignment gives ψ under

$$g(a[z \mapsto u]) = (ga)[z \mapsto gu].$$

Thus gu witnesses the existential formula under ga . Apply the same argument with g^{-1} for the converse. Universal quantification follows directly by the bijection on witnesses, or by negation and existential quantification. Bound variables are handled by these modified assignments, so the conclusion depends only on the displayed free variables. $\square$

A parameter-free formula in the pure equality language therefore defines only a union of equality-pattern classes. This concerns a fixed carrier: a sentence may distinguish carriers of different cardinalities without distinguishing atoms within one carrier. If parameters $p_1, \dots, p_k$ are held fixed, the same proof guarantees invariance under their pointwise stabilizer. Alternatively, transport the parameters together with the other inputs. A named atom cannot remain fixed while an arbitrary permutation moving it is still counted as a symmetry of the complete interpreted language.

Corollary 3.10 (Unique definition does not evade the selection criterion). *Under the lemma's hypotheses, if a formula $\varphi(\bar{x}, y)$ defines at most one y for each input tuple, its exact existence domain is invariant and the resulting partial function is equivariant. For a pure equality carrier with at least two atoms, no parameter-free formula with only y free uniquely defines an atom without input.*

Proof. Formula invariance is covariance of its functional result relation. Apply Theorem 5.2. With no input, a unique output would be fixed by every permutation, which no atom of a nonsingleton equality-only carrier is. $\square$

The lemma licenses arbitrarily nested finite formulas built from the stated primitives. It does not assert that every meaningful operation has such a formula, or that semantic invariance implies first-order definability. General conservation still follows from the actual original-incidence warrants. A broader grammar requires its own preservation proof; introducing a new primitive with an unproved transport law does not inherit this one.

3.7 What the pre-metric argument establishes upstream

The dependency is now explicit: an actual incidence supplies the conditions of correct attribution; those conditions distinguish a warranted target relation from a bare representative name; equal-content presentations cannot acquire discordant authority; and a complete equality-only interface therefore permits the full relabeling group. Its structural dimensionlessness follows by the exact orbit and stabilizer calculations. A richer retained relation changes the relevant symmetry problem, while still incurring the same kernel and the same demand for original-incidence warrant.

The construction does not infer non-action from absence of a metric. Identity maps, coordinate projections, equality tests, and invariant class codes already show why that inference

would fail. It instead locates a specific dependency problem for any proposed boundary operation: the operation must not use a target association or interior licence as the unsupported source of that very association or licence. Establishing the corresponding non-transmission and no-self-licensing results requires the separate analysis of retained information and warrant dependencies. This pre-fixational sequence supplies their constitutive starting point without treating a logical boundary as a physical wall, a temporally earlier world, or an exemption from governance.

4 Invariant information and lawful selection

An admissible redescription carries the data of an incidence into another presentation while retaining the bearer, complete question, and conditions relevant to the answer. The kernel requires that this retention have a determinate meaning. The mathematical question is then exact: which observations and selections can commute with the independently established redescriptions? The answer depends on their actual action. No assumption of full permutation symmetry is needed in this section.

4.1 The information retained by an action

Definition 4.1 (Presentation action and invariant observation). Let a group G act on a set X of presentations. Write

$$[x]_G = \{gx : g \in G\}, \quad \pi : X \longrightarrow X/G, \quad \pi(x) = [x]_G.$$

For a set V with the trivial G -action, an observation $f : X \rightarrow V$ is *invariant* when $f(gx) = f(x)$ for all g, x . Here “scalar” means that the output is compared without a further transformation; it need not be a real number.

The relevant action is specified from the data and their lawful transport, before selecting the answer to be conserved. Taking G to consist of the transformations that happen to preserve a proposed answer would establish only a property of that definition. An application instead identifies why those transformations preserve the actual incidence and its warrants.

Theorem 4.2 (Exact invariant-information boundary). *An observation $f : X \rightarrow V$ is invariant if and only if there is a unique function $\bar{f} : X/G \rightarrow V$ such that*

$$f = \bar{f} \circ \pi.$$

Thus X/G retains exactly the information expressible by invariant observations with arbitrary set codomains.

Proof. If f is invariant, set $\bar{f}([x]_G) = f(x)$. If $[x]_G = [y]_G$, then $y = gx$ for some g , so the proposed values agree. The definition is therefore well defined. Since π is surjective, the factorization determines $\bar{f}$ uniquely. Conversely, a factorization gives $f(gx) = \bar{f}([gx]_G) = \bar{f}([x]_G) = f(x)$. Finally, the orbit map itself is invariant. Consequently any two distinct orbits can be distinguished by an invariant observation, namely π ; no additional identifications are forced by invariance alone. $\square$

Corollary 4.3 (Constancy on a transitive presentation space). *If X is nonempty and the action is transitive, every invariant observation on X is constant. More generally, an invariant observation may distinguish different orbits and is constant on each orbit.*

Proof. A transitive action has a singleton orbit quotient. The general statement is the factorization in Theorem 4.2. $\square$

The theorem constrains information by its role under transformation, rather than by its notation. A real number and a discrete index obey the same factorization criterion. Neither can distinguish two presentations in one orbit as an invariant answer. Both may distinguish different orbits when the data provide that distinction. Bivalence of an output places no further restriction on the number of input orbits.

Remark 4.4 (Covariant outputs). When G also acts on an output set Y , the appropriate condition can instead be *equivariance*, $s(gx) = gs(x)$. A vector's components, a named endpoint, or a chosen tuple entry may change together with its presentation. Such an output need not define a function $X/G \rightarrow Y$ with unchanged values. One must either compare transported outputs in a common frame, pass to an appropriate output quotient, or retain the equivariant map. The scalar theorem does not prohibit a covariant realization.

4.2 The stabilizer test for lawful selection

A selection must produce an eligible output for the given input. This requirement distinguishes selection from an arbitrary map into an ambient set containing irrelevant fixed points. The stabilizer test asks whether an eligible output survives every lawful transformation that leaves its input fixed; the following theorem states both its necessity and its precise existence converse.

Definition 4.5 (Eligible family and lawful partial selector). Let G act on X and Y , and let $D \subseteq X$ be G -invariant. An *eligible family* on D assigns a subset $F(x) \subseteq Y$ to each $x \in D$, with

$$F(gx) = gF(x).$$

An equivariant selector on this family is a function $s : D \rightarrow Y$ satisfying $s(x) \in F(x)$ and $s(gx) = gs(x)$. Relative to X , it is a partial selector whose domain is preserved by redescription. Define

$$G_x = \{g \in G : gx = x\}, \quad \text{Fix}_Y(G_x) = \{y \in Y : (\forall h \in G_x) hy = y\}.$$

Theorem 4.6 (Eligible fixed-point criterion). *Every equivariant selector satisfies*

$$s(x) \in F(x) \cap \text{Fix}_Y(G_x) \quad (x \in D).$$

In particular, an empty intersection at any input prevents such a selector on D .

Conversely, suppose an orbit representative x_i has been supplied for each orbit in D , together with an element $y_i \in F(x_i) \cap \text{Fix}_Y(G_{x_i})$. Then

$$s(gx_i) = gy_i$$

defines an equivariant selector. Every equivariant selector is recovered from its values on these representatives.

Proof. For necessity, eligibility gives $s(x) \in F(x)$. If $h \in G_x$, then $hs(x) = s(hx) = s(x)$, proving the fixed-point condition. For the converse, suppose $gx_i = hx_i$. Then $h^{-1}g \in G_{x_i}$, so $gy_i = hy_i$. Each input belongs to exactly one represented orbit, and the formula is therefore well defined. Since $gy_i \in gF(x_i) = F(gx_i)$, it selects an eligible output. For $k \in G$, the same formula gives $s(kgx_i) = kgy_i = ks(gx_i)$, proving equivariance. Finally, equivariance forces any selector's value at gx_i to be the transport of its value at x_i . $\square$

The necessity direction requires neither a choice principle nor an algorithm. The converse explicitly displays the choices used. For an arbitrary family of orbits, nonemptiness of the eligible fixed-point sets yields a global selector under the corresponding choice principle; it does not provide a canonical or effective selector. With finitely many orbits, the required finite choices suffice. These are existence statements, not claims about physical attainability or computational cost.

Corollary 4.7 (A symmetric two-branch obstruction). *Suppose $F(x) = \{a, b\}$ with $a \neq b$, and some $h \in G_x$ exchanges a and b . No equivariant selector on an invariant domain containing x can select from this family.*

Proof. Neither eligible output is fixed by h , so $F(x) \cap \text{Fix}_Y(G_x) = \emptyset$. Apply Theorem 4.6. $\square$

Example 4.8 (Selecting one indistinguishable token). An input records two eligible tokens and the relation that they occupy symmetric positions. Exchanging the tokens preserves the entire input and exchanges the two possible answers to “which token?” The corollary excludes an equivariant singular answer from those data. An unordered answer containing both tokens is a different output type. An independently warranted mark can instead distinguish an output, but then the input contains the mark and has a different stabilizer. The original obstruction is neither repaired nor contradicted by that new incidence.

Invariance and selection therefore impose distinct tests. An invariant answer is constant along presentation orbits. A selected object must also survive every symmetry that leaves its input fixed. The latter is a stabilizer condition, so even a nontransitive action can prevent selection at a particular input. If output identity is itself represented by an equivalence relation, the action and eligible family must first be shown to descend to that identity quotient; distinctness in the obstruction then means distinct output identities.

The invariant-domain clause prevents a final evasion. Defining a partial selector at one arbitrarily chosen presentation, while leaving its redesciptions outside the domain, can make an “equivariant whenever both sides are defined” test vacuous. Such a domain already performs the unaccounted selection. Definition 4.5 includes definedness among the facts that lawful redescription preserves.

4.3 What additional fixation must supply

The obstruction theorem can also characterize the work of an additional datum. Let G act on a set Z of possible anchor data, and let $W \subseteq D \times Z$ be an invariant set of licensed input–anchor pairs. The candidate family at (x, z) remains $F(x)$ unless the new datum also carries an independently justified eligibility refinement. The action on W is diagonal.

Theorem 4.9 (Exact anchor criterion). *An equivariant selector on W must satisfy*

$$s(x, z) \in F(x) \cap \text{Fix}_Y(G_x \cap G_z).$$

Conversely, supplied orbit representatives of W and eligible elements of these fixed-point intersections determine every such selector by transport. In particular, an anchor that leaves the input stabilizer unchanged cannot remove a fixed-point obstruction present at x .

Proof. The stabilizer of (x, z) in the diagonal action is exactly $G_x \cap G_z$. Apply Theorem 4.6 to the eligible family $(x, z) \mapsto F(x)$. If the intersection equals G_x , its eligible fixed-point set is unchanged. $\square$

This makes fixation a checkable operation rather than an unexplained instruction to choose. A record tying a serial number to the tested battery can reduce the relevant stabilizer; a fresh decorative number with no warranted connection to either battery does not. Whether an anchor is licensed belongs to its own original incidence. The group calculation identifies the discrimination it supplies once that connection has been established.

Proposition 4.10 (Finite lower bound on discriminating anchor states). *Let Y be a finite transitive G -set with $|Y| = m > 0$, and let A be a nonempty G -set of anchor states admitting an equivariant selection $s : A \rightarrow Y$. Every orbit $O \subseteq A$ maps surjectively onto Y . Hence $|O| \geq m$; if O is finite, its cardinality is a positive multiple of m . The bound is attained by $A = Y$ and $s = \text{id}_Y$.*

Proof. For $a \in O$ and any $y \in Y$, transitivity supplies g with $gs(a) = y$, and then $s(ga) = y$. Thus $s|_O$ is onto. For $y, y' \in Y$, a group element sending y to y' maps the corresponding fibres in O bijectively, with inverse supplied by its inverse element. All finite fibres therefore have the same positive cardinality, and $|O| = m|s^{-1}(y) \cap O|$. The displayed identity map establishes attainability. $\square$

The lower bound counts distinguishable anchor states; it does not grade standing or assign a physical cost. The attaining example already supplies the selected identity as anchored data. It shows how much discrimination suffices, not how an unwarranted choice acquires a warrant. For two symmetric batteries, a datum unchanged by their exchange cannot supply the missing identification, while a warranted two-way identification can. The argument remains the same for any finite number of symmetric alternatives.

4.4 Randomized selection and a change of output type

A probabilistic procedure is a serious test of a no-selection claim. It can treat symmetric candidates equally while returning one candidate in a particular run. The distinction between its probability law and its realized output must therefore be made inside the mathematics.

Proposition 4.11 (Invariant finite laws and selected identities). *Let a group H act on a nonempty finite candidate set B . A probability mass function $\mu : B \rightarrow [0, 1]$ is invariant under H exactly when it is constant on each H -orbit. Such laws are precisely the assignments of a total mass $w_O \geq 0$ to each orbit O , with $\sum_O w_O = 1$, distributed uniformly within that orbit. In particular, a transitive action has exactly one invariant law, the uniform law. An invariant law is a point mass exactly when its support is one H -fixed candidate.*

Proof. Invariance says $\mu(hb) = \mu(b)$, which is equivalent to constancy on the orbits. If O has n_O elements and total mass w_O , each of its points therefore has mass w_O/n_O . Conversely these assignments are nonnegative, have total mass one, and are invariant. For a transitive action there is one orbit with mass one. The support of a point mass is preserved exactly when its unique point is fixed by every element of H . $\square$

At an input x in Theorem 4.6, take $H = G_x$ and a nonempty finite eligible set $B = F(x)$. Even when B has no fixed point, its uniform law is invariant under the input stabilizer. Indeed, the assignment of the uniform law to every such $F(x)$ is an equivariant distribution-valued output, because the group action transports $F(x)$ bijectively to $F(gx)$. It is a lawful mathematical answer to the question asking for that distribution. It is not a fixed element of $F(x)$.

A realized sample adds a further incidence: the outcome of the trial and its connection to the selected candidate. A deterministic description of that run must include the relevant

outcome or random input and its actual transformation law. The anchor theorem then tests the enlarged input. In particular, a coin outcome left unchanged by exchanging two batteries cannot by itself determine which battery it names; the sampling apparatus must also establish that association. A random procedure can choose a battery for a new task without discovering which original battery passed the earlier trial.

There is a useful exact counterpart when the output is offered as an answer to an already determinate question. Suppose $J(x, v)$ has the unique warranted result v_x . For a probability law on a finite output set containing v_x , a claim to give a warranted answer with probability one requires

$$\mu_x(\{v : J(x, v)\}) = 1,$$

and hence is the point mass at v_x . This statement concerns the output law. Warranting an actual reported result still concerns its original incidence; a probability-one assertion by itself does not identify or validate every individual run. A nontrivial probability of other values describes a fallible procedure, a different question, or a different incidence relation; it is not a second warranted answer to the unchanged determinate question. This does not exclude stochastic physical laws. A distribution can itself be the determinate object of a query, while a future individual outcome remains unfixed. The theorem constrains the claimed output and its original warrant in each case.

5 From original incidences to unique lawful selection

Same-incidence conservation compares warranted answers after their bearer and complete question have been identified. An equivariance theorem addresses a further issue: how are answers represented when their input presentation changes? The connecting premise is transport of the independently specified result relation. This section states that premise exactly, derives the domain and output laws from it, and distinguishes existence of a lawful selector from its uniqueness. The final formulation allows different presentations to have different output types.

5.1 Transporting an original result relation

Definition 5.1 (Covariant functional incidence). Let a group G act on sets X and Y . The elements of X encode the complete input incidences relevant to the question, and Y contains their possible represented results. Fix a relation $J \subseteq X \times Y$ independently of any proposed consumer. Assume partial functionality:

$$J(x, y) \wedge J(x, z) \implies y = z. \quad (2)$$

Its exact incidence domain is

$$D_J = \{x \in X : \exists y \in Y \ J(x, y)\}.$$

The original relation is covariant under the specified actions when

$$J(gx, gy) \iff J(x, y) \quad (g \in G, x \in X, y \in Y). \quad (3)$$

The actions and this equivalence must be warranted from the original incidence conditions, including the transformation of every relevant parameter.

Theorem 5.2 (Original-incidence transport determines equivariance). *For a partially functional relation J , let $f_J : D_J \rightarrow Y$ assign the unique incident result. Then (3) holds if and only if D_J is G -invariant and f_J is equivariant. In particular, a covariant original relation determines exactly one function with that graph, including its exact domain.*

Proof. For each $x \in D_J$, existence and (2) give exactly one result; this defines f_J without making choices among alternatives. If $J(x, y)$, covariance gives $J(gx, gy)$, so $gD_J \subseteq D_J$. Applying the same argument to g^{-1} gives equality. For $x \in D_J$, both $f_J(gx)$ and $gf_J(x)$ are incident at gx . Functionality therefore gives

$$f_J(gx) = gf_J(x).$$

Conversely, $J(x, y)$ holds exactly when $x \in D_J$ and $y = f_J(x)$. Domain invariance and equivariance transform this assertion into $gx \in D_J$ and $gy = f_J(gx)$, which is $J(gx, gy)$. Using the inverse action proves the equivalence. A function reproducing the entire graph has precisely this domain and these unique values. $\square$

Covariance does not assert $gx = x$ or $gy = y$: it transports the complete input and its result together. Two complete configurations may be exchanged even though the original answer at each is determinate. A coarse available-data interface may conceal that difference. Consequently a symmetry fixing the available data need not fix the complete configuration. A selection obstruction must use the action and eligible outputs of the selector's actual input space; ignorance of an existing distinction does not erase it from the original incidence. When the action on Y is trivial, this yields an invariant scalar answer and hence its unique factorization through D_J/G . In the general case the answer is covariant. If a total outcome map is desired, adjoin a fresh tag $\perp$, let every g fix it, and put $\hat{f}_J(x) = \perp$ outside D_J . The theorem makes $\hat{f}_J : X \rightarrow Y \sqcup \{\perp\}$ equivariant. Here the absent tag means non-incidence in the original relation. It does not mean that a particular search has timed out or failed to find a proof.

A reader $r : C \rightarrow Y$, with $C \subseteq X$, whose outputs satisfy $J(x, r(x))$ agrees with f_J on C , but this alone neither identifies C with D_J nor makes C invariant. A reader defined at only one of two interchangeable presentations illustrates the gap. Invariance of C is a separate requirement for an equivariant partial implementation; completeness additionally requires the claimed coverage of D_J .

The theorem is an exact bridge, not an independent proof of its covariance premise. Taking J to be the graph of the very function whose equivariance is sought merely restates the desired conclusion. The substantive route specifies the original equations, incidence rules, or physical conditions first, then proves their transport law. Choosing a group merely to preserve a preferred output does not establish that it exhausts the relevant lawful comparisons.

Example 5.3 (Solutions under change of basis). Fix $n \geq 1$, let $G = \text{GL}(n, \mathbb{R})$, let X consist of pairs (A, b) with A invertible and $b \in \mathbb{R}^n$, and let $Y = \mathbb{R}^n$. Put

$$g(A, b) = (gAg^{-1}, gb), \quad gy = gy, \quad J((A, b), y) \iff Ay = b.$$

Invertibility establishes functionality and existence. The independent equation gives

$$(gAg^{-1})(gy) = gb \iff Ay = b.$$

The theorem therefore derives equivariance of the unique solution; it was not postulated of a desired answer. This example concerns simultaneous change of basis. If a metric, boundary condition, or distinguished vector belongs to the complete problem, its coordinates must also be transported. The calculation does not license suppressing that structure.

5.2 An exact uniqueness criterion for eligible selectors

Let G act on X and Y , let $D \subseteq X$ be invariant, and let $F(x) \subseteq Y$ satisfy $F(gx) = gF(x)$ on D . An eligible equivariant selector is a function $s : D \rightarrow Y$ satisfying $s(x) \in F(x)$ and

$s(gx) = gs(x)$. Define

$$K_x = F(x) \cap \text{Fix}_Y(G_x), \quad G_x = \{g \in G : gx = x\}.$$

Every such selector takes its value at x in K_x . More precisely,

$$K_{gx} = gK_x. \tag{4}$$

Indeed, $G_{gx} = gG_xg^{-1}$; a point fixed by G_x is carried by g to one fixed by G_{gx} , eligibility transports by hypothesis, and g^{-1} proves the reverse inclusion.

Theorem 5.4 (Unique eligible selection). *There exists exactly one eligible equivariant selector on D if and only if K_x is a singleton for every $x \in D$. This equivalence requires no axiom of choice. If D is empty, both assertions hold: the pointwise condition is vacuous and the empty function is the unique selector.*

Proof. Suppose first that every K_x is a singleton. Assign $s(x)$ its unique element. Unique specification defines a function in ordinary set theory without an axiom selecting among multiple elements. By (4), $gs(x)$ belongs to K_{gx} , so uniqueness there gives $s(gx) = gs(x)$. The function is eligible by construction. Every other eligible equivariant selector has its value in the same singleton at each input, so equals s .

Conversely, suppose s is the unique eligible equivariant selector. Its value makes each K_x nonempty. Fix any $x \in D$ and any $z \in K_x$. Define a replacement on the orbit of x by

$$s_z(gx) = gz,$$

leaving $s_z(u) = s(u)$ outside that orbit. This rule is well defined: if $gx = hx$, then $h^{-1}g \in G_x$, so $gz = hz$. For every point of the orbit some such g exists and all such g give the same result; no family of choices of group elements is needed. Eligibility follows from $gz \in gF(x) = F(gx)$. The orbit and its complement in D are invariant, and the transport rule proves equivariance on the orbit, while the old selector does so on the complement. Thus s_z is another eligible equivariant selector. Uniqueness implies $z = s_z(x) = s(x)$. Since z was arbitrary, $K_x = \{s(x)\}$. $\square$

The necessity argument uses the selector whose existence is part of the uniqueness assertion. It does not infer a global selector merely because every K_x is nonempty. General existence still requires compatible choices across the input orbits, as in the orbit-representative construction. The singleton criterion removes that choice burden and proves more than existence: the entire selector is forced by the supplied structure.

This is mathematical uniqueness relative to the stated candidate family. A family describing necessary conditions for a physical endpoint need not already characterize physically realized endpoints. Even a unique mathematical selector requires the independent incidence or realization warrant that identifies its selected values with the original physical question. Conversely, when the candidates are the actual results of the covariant functional relation above, that warrant and uniqueness are supplied by the original relation itself. Neither route permits agreement of desired outputs to substitute for the identification of the bearer and complete question.

5.3 Different presentation types: a groupoid formulation

A single group action assumes one ambient input set and one ambient output set. Coordinate charts, encodings, and typed descriptions can instead have different output sets joined by reversible comparisons. The appropriate generalization is a small groupoid: a category $\mathcal{C}$

whose objects and arrows form sets and whose every arrow is invertible. Two objects are in the same connected component when an arrow joins them; inverse and composition make this an equivalence relation.

Definition 5.5 (Transported output family). For each object c give a set V_c , and for every arrow $\alpha : c \rightarrow d$ give a bijection $T_\alpha : V_c \rightarrow V_d$ such that

$$T_{\text{id}_c} = \text{id}_{V_c}, \quad T_{\beta\alpha} = T_\beta T_\alpha.$$

Thus the output family is a functor from $\mathcal{C}$ to sets. Give eligible subsets $E_c \subseteq V_c$ with $T_\alpha(E_c) = E_d$. A coherent eligible section is a family $s_c \in E_c$ satisfying $T_\alpha(s_c) = s_d$ for every arrow. Its eligible loop-fixed set at c is

$$L_c = \{v \in E_c : T_\ell(v) = v \text{ for every loop } \ell : c \rightarrow c\}.$$

Loops here are arrows returning to one presentation, not assumed to be identities. Their action records the obstruction to a result being independent of the comparison route. Conjugation by an arrow gives

$$T_\alpha(L_c) = L_d \quad (\alpha : c \rightarrow d). \quad (5)$$

To see this, a loop ℓ at d conjugates to $\alpha^{-1}\ell\alpha$ at c ; apply functoriality and then use α^{-1} for the reverse inclusion.

Theorem 5.6 (Coherent sections from component data). *Suppose one object c_i has been supplied in each connected component, indexed by a set I . Evaluation at these representatives is a bijection between coherent eligible sections and*

$$\prod_{i \in I} L_{c_i}.$$

In particular, supplied values $v_i \in L_{c_i}$ determine a unique coherent section. No additional choices of connecting arrows are required.

Proof. A coherent section is fixed by every loop, so evaluation gives a member of the displayed product. Conversely, for c in the component of c_i , define $s_c = T_\alpha(v_i)$ for any arrow $\alpha : c_i \rightarrow c$. Such an arrow exists. If β is another, then $\beta^{-1}\alpha$ is a loop at c_i , and therefore

$$T_\alpha(v_i) = T_\beta T_{\beta^{-1}\alpha}(v_i) = T_\beta(v_i).$$

The value at each c is consequently uniquely specified, without choosing arrows simultaneously. It is eligible by transport. For $\gamma : c \rightarrow d$, the composite $\gamma\alpha$ supplies $s_d = T_\gamma(s_c)$, proving coherence. Every coherent section extending v_i has these values, so evaluation and construction are inverse. $\square$

The representative family is supplied data; its existence is not asserted without the corresponding choice principle. Likewise, nonemptiness of all the factors does not by itself choose a point of an arbitrary product. A discrete groupoid already recovers the usual problem of choosing from an arbitrary family of nonempty sets. These are genuine existence burdens, whereas the following uniqueness theorem avoids them entirely.

Theorem 5.7 (Unique coherent section). *There exists exactly one coherent eligible section if and only if every L_c is a singleton. This includes the empty groupoid and requires no axiom of choice or supplied component representatives.*

Proof. If every L_c is a singleton, take its unique member. Equation (5) makes this family coherent, and any coherent eligible section must have those same values.

Conversely, let s be the unique coherent eligible section. Fix an object c and $v \in L_c$. On its component, transport v along arrows out of c ; the loop argument in the preceding proof gives unique values without choosing arrows. On every other component retain s . The resulting section is eligible and coherent, since no arrow joins different components. Uniqueness makes its value v at c equal to s_c . Since $s_c \in L_c$, this proves that L_c is a singleton. When there are no objects the unique section is the empty family. $\square$

Corollary 5.8 (Typed original-incidence transport). *Suppose an independently specified result predicate $J_c(v)$ on each V_c has at most one solution and satisfies*

$$J_d(T_\alpha v) \iff J_c(v) \quad (\alpha : c \rightarrow d).$$

The objects where a result exists form a union of connected components. On their full subgroupoid, the incident results form the unique coherent section eligible in $E_c = \{v : J_c(v)\}$.

Proof. An incident result transports along an arrow to an incident result, and the inverse arrow proves the converse existence implication. Each nonempty E_c is a singleton by functionality; transport makes its unique member coherent. Every loop fixes this member, so $L_c = E_c$, and the uniqueness theorem applies. $\square$

For an action groupoid, objects are inputs $x \in D$, an arrow labelled g goes from x to gx , and output transport is the given action on Y . Its loops at x are exactly the elements of G_x . The groupoid results therefore recover the preceding selector theorems while allowing varying result types and more general reversible comparisons. They do not automatically apply to noninvertible updates. In every version the transport laws express preservation of independently identified incidences; they do not create a physical relation, certify an incomplete candidate family, or establish a desired dynamics by symmetry alone.

6 The complete equality-only boundary

The general theorems admit an exact specialization when equality is the only authoritative relation on the presented atoms. This specialization identifies everything such an interface can express, including its remaining relational information. An absence of named or preferred atoms does not by itself establish this hypothesis: an unmarked cycle, for example, retains an adjacency relation. Its lawful permutations are those preserving that relation, and Section 4 applies with that smaller group.

Definition 6.1 (Equality-only interface). Let P be a nonempty set whose authoritative atom structure is equality. Its automorphism group is $\text{Sym}(P)$, acting on finite tuples by

$$g(x_1, \dots, x_n) = (gx_1, \dots, gx_n).$$

The coordinate positions are fixed. For $n \in \mathbb{N}$, including $n = 0$, write $[n] = \{1, \dots, n\}$ and let $\text{Eq}(x)$ be the partition of $[n]$ defined by $i \sim_x j$ exactly when $x_i = x_j$. Let $\mathcal{E}_n(P)$ denote the set of partitions realized by tuples in P^n .

All bijections preserve and reflect equality. Hence the full symmetric group follows from the stated equality-only structure, rather than from the bare fact that no element has been selected. An application must establish which available data have this structure. The atom permutations do not permute coordinate positions; a position such as “first entry” can be part of the input without giving an atom an intrinsic name.

6.1 Orbit classification and invariant observations

Lemma 6.2 (Extension of finite correspondences). *Every bijection $u : A \rightarrow B$ between finite subsets of P extends to a permutation of P that fixes every element outside $A \cup B$.*

Proof. Put $U = A \cup B$. Since $|A| = |B|$, the finite sets $U \setminus A$ and $U \setminus B$ have equal cardinality. Choose a bijection between them and combine it with u to obtain a permutation of U . Extend this permutation by the identity on $P \setminus U$. $\square$

Theorem 6.3 (Equality-pattern completeness). *For finite n , two tuples in P^n lie in the same $\text{Sym}(P)$ -orbit if and only if they have the same equality partition. Consequently, every invariant function $f : P^n \rightarrow V$, with V carrying the trivial action, has a unique factorization*

$$f = \bar{f} \circ \text{Eq}, \quad \bar{f} : \mathcal{E}_n(P) \rightarrow V.$$

Conversely, every such $\bar{f}$ defines an invariant observation.

Proof. A permutation preserves equality, so it preserves the partition. Conversely, if $\text{Eq}(x) = \text{Eq}(y)$, the rule $x_i \mapsto y_i$ is well defined and bijective between the sets of entries occurring in the two tuples. Lemma 6.2 extends it to a permutation carrying x to y . The factorization and its uniqueness now follow from Theorem 4.2, since Eq maps onto the realized partitions. $\square$

A partition is realized exactly when its number of blocks is at most the cardinality of P . Thus small carriers cause no exceptional gap in the theorem. For $n = 0$, P^0 contains one empty tuple and $\mathcal{E}_0(P)$ contains the empty partition; the result applies unchanged. For $n = 1$, there is one pattern, so every invariant scalar observation is constant. For $n = 2$ and $|P| \geq 2$, equality and inequality are the two patterns. In general the theorem classifies all invariant relations by taking $V = \{0, 1\}$ and all other scalar observations by their chosen codomains.

Example 6.4 (Bivalent output with relational input). Suppose P has at least three atoms. For $x \in P^3$, let $k(x)$ be the number of distinct entries, and put $a(x) = 1$ precisely when $k(x) \geq 2$. Both functions are invariant. The positive patterns with $k = 2$ and $k = 3$ remain distinct input patterns; they are not extra standing values. This example classifies a possible binary predicate. Whether that predicate answers a particular admission question requires the question's independent warrant.

Proposition 6.5 (Order and metric consequences). *If $|P| \geq 2$, no asymmetric binary relation invariant under $\text{Sym}(P)$ relates two distinct atoms. Every real-valued invariant metric on P has the form*

$$d_c(x, y) = \begin{cases} 0, & x = y, \\ c, & x \neq y, \end{cases} \quad c > 0.$$

Conversely, every d_c is an invariant metric.

Proof. If a relation holds of distinct x, y , their transposition makes it hold of y, x , contradicting asymmetry. For the metric, the diagonal is zero by the metric axioms. Theorem 6.3 makes its value constant on all unequal pairs, and separation makes this value positive. Conversely, d_c is symmetric and separates points. If $x \neq z$, at least one of $x \neq y$ or $y \neq z$ holds, so $d_c(x, z) \leq d_c(x, y) + d_c(y, z)$. Invariance follows from preservation of equality. $\square$

Thus no invariant strict total order exists when there are at least two atoms. A metric does exist, but has no distance geometry beyond equality. Symmetry leaves its positive scale c undetermined: selecting units or a normalization requires an additional convention

or warrant. For a singleton carrier the only metric is zero and there is no off-diagonal scale. These statements express the complete information restriction; they do not claim that writing a number or a metric is itself forbidden.

6.2 Complete classification of equivariant atom outputs

Scalar constancy does not apply to an output atom that transforms with the input. In particular, a coordinate projection is equivariant. There are further possibilities, and their exact classification prevents an overstatement of the selection prohibition.

Lemma 6.6 (Fixed atoms of a tuple stabilizer). *For $x \in P^n$, put $A_x = \{x_i : i \in [n]\}$. Then*

$$\text{Fix}_P(\text{Sym}(P)_x) = \begin{cases} A_x \cup \{u\}, & P \setminus A_x = \{u\}, \\ A_x, & |P \setminus A_x| \neq 1. \end{cases}$$

Proof. A permutation fixes the tuple exactly when it fixes every member of A_x individually. It may permute the complement arbitrarily. Every used atom is therefore fixed. If the complement is a singleton, its unique atom is fixed as well. If the complement contains at least two atoms, a transposition moves each chosen unused atom while fixing the tuple. An empty complement contributes no additional atom. $\square$

Theorem 6.7 (All equivariant finite-tuple outputs). *Equivariant functions $s : P^n \rightarrow P$ are in bijection with the following rules on the realized equality partitions. For each $\lambda \in \mathcal{E}_n(P)$, choose either:*

1. *a block of λ , and output the common atom occupying its positions; or*
2. *the unused-atom option, available exactly when tuples with pattern λ leave one atom of P unused, and output that unique atom.*

The rule must assign an available option to every realized partition.

Proof. By Theorem 4.6, with eligible family $F(x) = P$, any equivariant output is fixed by the tuple stabilizer. Lemma 6.6 gives precisely the listed possibilities. If two tuples have pattern λ , a permutation carries one to the other. It carries the atom of each coordinate block to the atom of that same block, and carries a unique unused atom to the unique unused atom. Equivariance therefore forces the option to depend only on λ . Conversely, any rule on partitions defines an output by the displayed options. Permutations preserve the partition and transport each option as described, proving equivariance. Distinct options give distinct atoms on every tuple realizing that partition, which proves uniqueness of the rule. $\square$

For $n \geq 1$, a block is always available. A rule may select different blocks on different equality patterns, so it need not be one fixed coordinate projection. If P is infinite, or finite with $|P| \geq n + 2$, there is never a unique unused atom and every equivariant output is an entry of the tuple. For a two-element P and $n = 1$, returning the other atom is equivariant and is not a projection. For $n = 0$, the theorem gives a function $P^0 \rightarrow P$ exactly when P is a singleton. All these cases follow from the same stabilizer test.

Corollary 6.8 (No selection from an unmarked nonsingleton set). *Let $\mathcal{D}$ be a $\text{Sym}(P)$ -invariant family of nonempty subsets of P . There is an equivariant function $c : \mathcal{D} \rightarrow P$ with $c(B) \in B$ if and only if every member of $\mathcal{D}$ is a singleton. In particular, when $|P| \geq 2$, there is no equivariant selector on all nonempty subsets and no invariant choice of an atom without input.*

Proof. If B contains at least two atoms, each candidate $b \in B$ is moved by a transposition within B that fixes B as a set. Thus the eligible fixed-point intersection is empty. If every B is a singleton, returning its unique member defines an equivariant selector. An input-free atom would have to be fixed by every permutation, which no atom is when $|P| \geq 2$. $\square$

The equality-only boundary is consequently exact in both directions. It excludes distinctions that the input cannot support and explicitly retains every invariant pattern and every equivariant output that it can support. Its necessity concerns authority from the stated interface. The complete question and independently warranted evidence determine which of these mathematically available functions, if any, supplies the standing-bearing answer.

7 The pre-metric non-transmissive boundary and constructed Axiom 6

The pre-fixational argument concerns the source of a distinction's authority. The equality-only theorems determine exactly what is available before any richer target-fixing structure has been independently licensed. A further question is whether structure used after fixation can be sent backwards to supply its own missing license. This is the non-transmission question. It is central to the AMetric boundary and cannot be replaced by a statement about scalar invariance alone.

The boundary is an order of constitutive dependence. A result must have the incidence that qualifies it; an operation's standing-bearing application must meet its domain and use conditions. A description of the operation does not furnish those conditions merely by describing a successful application. These relations need not be evaluated in an earlier physical event. Actual performance can establish that an independently specified test was met. What cannot establish qualification is the circular assertion that the performance was licensed because it is being treated as licensed.

The qualification question itself has an actual determinate bearer, such as a proposal record or a putative operation application. The target identity that the proposal seeks to authorize may remain unfixed within its retained content. These are different levels: a determinate proposal need not already identify the target it claims to select.

Every actual proposal, mark, apparatus, mathematical model, and resulting object already incurs kernel governance. Pre-fixation concerns the absence of a particular licensed target discrimination within the proposal content. It is not a region of physically existing things exempt from the kernel. Likewise, "pre-metric" concerns priority to the use of a metric as authoritative structure, not an unproved claim that a physical universe once lacked geometry.

7.1 The fibre test and the direction of non-transmission

There are three different operations that ordinary language can call a transfer. First, one can copy or encode a description. Second, one can prove that a source witness supports a target claim. Third, one can purport to let a target claim authorize the very witness or operation whose license it presupposes. The first is not by itself the second. The third cannot replace the independent warrant required by the second.

A later metric, ordering, record, or operator can be an entirely legitimate construction. It can also disclose or measure an original distinction that was already present. Neither fact allows its target-specific authority to become a primitive premise of the unchanged proposal interface. If the interface contains the distinction needed to support its answer, that support

must be exhibited. If it does not, additional licensed data must enter the argument. If the proposed support consists only in the answer's asserted authority, the dependence is circular.

Definition 7.1 (Refinement and backward determination). Let X be a set of source presentations and Z a set of richer presentations. A map $p : Z \rightarrow X$ specifies exactly which original data are retained when the additional structure is forgotten. Let $w : Z \rightarrow V$ be a proposed target answer. A *backward determination from the retained source data* is a function $\bar{w} : \text{im } p \rightarrow V$ such that $w = \bar{w} \circ p$, where p is corestricted to its image. To claim such determination on all of X requires coverage $\text{im } p = X$ or an independently justified extension outside that image.

The definition specifies determination, not warrant. An application must independently establish what p forgets, what original question w answers, and why that answer has its claimed standing. Coding a relation as an element of V , instead of calling it a number, does not alter the issue. Values and dependent output types must first be compared by their justified identifications, as in Theorem 5.2 and corollary 5.8.

Theorem 7.2 (Exact non-transmission of additional discriminatory authority). *A backward determination exists if and only if*

$$p(z) = p(z') \implies w(z) = w(z') \quad (z, z' \in Z). \quad (6)$$

When it exists it is unique on $\text{im } p$. Consequently, if two richer realizations have the same retained source data but require different target answers, no function of those source data supplies the answer on both. If the answer does descend, the richer realization introduces no additional distinction into that answer at the source interface.

Proof. If $w = \bar{w} \circ p$, equal p -values give equal w -values. Conversely, define $\bar{w}(x)$ to be the common value $w(z)$ for $p(z) = x$. There is such a z for every $x \in \text{im } p$, and (6) makes the value independent of the representative. This specifies a function without choosing a representative for every fibre. The same equation forces uniqueness because every element of $\text{im } p$ is attained. A pair violating the condition contradicts any proposed factorization. The final statement means exactly that all distinctions relevant to w are already determined by p ; it does not assert that p determines the entire richer presentation. $\square$

This theorem is not a prohibition on an arbitrary physical channel. It is the exact obstruction to treating structure forgotten by p as a new source of authoritative discrimination at the unchanged source interface. No assumptions of distance, time, continuity, finite cardinality, or computation occur in its proof. Section 8 extends the same factorization test to the complete family of uses and its canonical maximal quotient.

For the equality-only interface $X = P^n$, its complete invariant atom information is the equality pattern. Any scalar authority claimed to be available from that interface must therefore factor through Eq by Theorem 6.3. A post-fixational metric or rank that assigns different required answers to tuples with the same pattern cannot be read backwards as an original equality-only discriminator. If it factors through the pattern, it adds no atom-specific discrimination there. If new metric data are independently licensed, the actual input has been enriched and its smaller lawful transformation group must be used. This is a precise division of cases, not a permission to call any unwelcome answer a change of scope.

The battery association. Let Z contain two complete arrangements, one in which battery a is associated with the passing original trial and another in which battery b is. Suppose p retains only the unassociated pair of names and the statement that one trial passed. The

answer naming the passing battery differs on the same p -fibre. An unattached “PASS” mark cannot transmit the missing association. A retained serial-number record can supply it, but the record’s association is additional independently warranted data. A later trial can supply a new result; it does not supply a different result for the original trial. Section 8.6 gives the full incidence and group-action model.

7.2 A license has no automatic export to another use

Source and target can both be determinate while the proposed transfer fails. Suppose $S_D : E_D \rightarrow \{0, 1\}$ and $S_F : E_F \rightarrow \{0, 1\}$ record two independently specified qualifications, and $h : E_D \rightarrow E_F$ is the proposed transfer. Positivity in D licenses positivity of the transferred target only when the original conditions establish

$$S_D(x) = 1 \implies S_F(h(x)) = 1. \quad (7)$$

This is a substantive incidence-preservation obligation. The equality of the two status alphabets supplies no proof of it.

For example, take $E_D = \{u, v\}$, let both source values be positive, and let the target qualifications of $h(u)$ and $h(v)$ differ. The single positive source bit cannot determine the target bit. This is a two-element instance of Theorem 7.2; it involves neither a third standing value nor an indeterminate physical object. A domain theorem establishing (7) makes the transfer lawful. A claim to a universal transfer must establish its quantifier over the complete relevant source domain.

For the same original locus, (7) can follow from independently warranted identification and conservation. For a new use, new operator domain, or different experimental question, the connecting theorem must be supplied. Necessary kernel governance is common to both domains; particular qualifications remain answerable to their particular incidences. Thus no operator, metric scale, or name exports its license merely by being copied into a new presentation.

7.3 The grounding test: an operation cannot supply its own first license

An operation may have a mathematically defined graph before a particular proposed application is shown to belong to its domain. A physical apparatus may act before anyone knows whether it meets a stated protocol. The missing object in either case is the warrant for the claimed application, not necessarily the map, apparatus, or event. The boundary’s operator-undefinedness claim concerns the operation *as a licensed application to that target*. Supplying the operator’s name, its intended type, or the result one expects from it does not establish its required domain incidence.

The grounding test gives an exact obstruction to self-licensing in grounded inference systems. It applies to finite certificates, but is not limited to them. An application must identify its independent grounds and justify its rules; the theorem below isolates the failure when every possible entrance already presupposes a member of the unseeded license family.

Definition 7.3 (Grounded rule closure). Let $\mathcal{J}$ be a set of judgments, $B \subseteq \mathcal{J}$ independently grounded judgments, and $\mathcal{R}$ a set of rules (A, j) with premise set $A \subseteq \mathcal{J}$ and conclusion $j \in \mathcal{J}$. A set $T \subseteq \mathcal{J}$ is rule-closed if $A \subseteq T$ and $(A, j) \in \mathcal{R}$ imply $j \in T$. Define

$$\text{Cl}_{\mathcal{R}}(B) = \bigcap \{T \subseteq \mathcal{J} : B \subseteq T \text{ and } T \text{ is rule-closed}\}.$$

The family being intersected is nonempty because it contains $\mathcal{J}$. An application claiming semantic warrant must separately prove the grounding of B and soundness of its rules.

Theorem 7.4 (No ungrounded self-authorization, including mutual licensing). *Let $C \subseteq \mathcal{J}$ be a family of proposed licenses. Suppose $B \cap C = \emptyset$ and every rule (A, j) with $j \in C$ has $A \cap C \neq \emptyset$. Then*

$$\text{Cl}_{\mathcal{R}}(B) \cap C = \emptyset.$$

This result requires neither finite premise sets nor a finite rule registry.

Proof. The complement $\mathcal{J} \setminus C$ contains B . It is rule-closed: if all premises of a rule lie in the complement, its conclusion cannot lie in C , since such a conclusion would require a premise in C . Hence the complement is one of the sets in the defining intersection. The closure is contained in it, proving the claim. $\square$

The theorem isolates the failure in a purported backward loan of authority. One proposed license depends on an operation declared legitimate only under that license; or several licenses support one another without any independent entrance into their dependency family. No chain of those rules supplies the absent ground. Writing the system as simultaneous equations does not solve this problem: an independently proved solution and its identification with the original qualification would provide new warrant, while merely repeating the circular assertions would not.

A rule with an independently warranted conclusion in C and premises outside C breaks the theorem's premise and can legitimately ground the license. An actual test can be such a source when its success conditions and its relation to the original bearer are established independently. The theorem does not demand a temporally earlier observer or forbid the very event being assessed from satisfying the protocol.

The distinction between grounded closure and mere fixed-point consistency matters. For the one-judgment rule $s \Rightarrow s$, both $\emptyset$ and $\{s\}$ are rule-closed, but closure of the empty ground contains no s . Choosing the larger closed set does not provide a warrant for s . Coinductive reasoning, cyclic proof systems with separately proved soundness conditions, and implicit mathematical definitions may be legitimate methods; their independent interpretation and soundness are the additional work. The theorem makes no unsupported claim that every semantic fact is enumerated by the displayed rule system.

7.4 Binary qualification and non-eventful fixation

Fix an original bearer and complete qualification question (b, q) . Let $A(b, q)$ be its qualification proposition, established from the domain's original conditions. Its classical status is

$$S(b, q) = \begin{cases} 1, & A(b, q), \\ 0, & \neg A(b, q). \end{cases}$$

The status alphabet has two elements. Both values are realized by a particular domain only if that domain supplies both a positive and a negative instance. Failure to obtain evidence is not a third qualification value and is not automatically evidence of the negative value. A proposed ill-typed question may fail to define this qualification problem at all.

Let a set I index putative implementations or stages of “crossing” at this unchanged complete locus, and suppose each supplies a separately warranted status S_i for that same original proposition. Then Theorem 2.4 gives

$$S_i = S(b, q) \quad (i \in I).$$

The status therefore cannot be advanced, graded, repaired, or changed by parametrizing its representation. This is true for continuous indices, discrete indices, and arbitrary index sets

alike. A process that gathers evidence can change a reader's knowledge; a new experiment can change which incidence is being assessed. Neither makes an independently warranted negative and positive answer coexist at the unchanged determinate locus.

In this precise sense, fixation at the boundary is *non-eventful*: the relation between a determinate incidence and its qualification is not an additional operation that manufactures its own qualification. A verification process has its own domain, evidence, and history. An operation that creates a new qualified object likewise has its own incidence. Its successful performance can ground the new qualification, but it cannot retroactively provide a different result for the original question or replace the conditions under which its own use was claimed to be licensed.

Binary standing does not mean that all positive objects, successful outputs, or reasons for qualification are identical. A bit can classify many different qualified records. Those distinctions remain part of the complete query family when intended uses require them. What is excluded is an additional *source of qualification authority* supplied solely by the index of a supposedly alternative crossing at the same locus. The equality-only no-individuator argument supplies a further selection obstruction where that index is supposed to choose an as-yet unfixed target.

7.5 The constructed boundary theorem

Sections 3–6 have supplied the pre-fixational argument and the exact information and selection classifications. The fibre and grounding theorems in this section establish the additional non-transmission claims. Theorem 7.5 now assembles these results: it is the transition from the upstream classification to the boundary that subsequent interface and construction arguments must respect. Those later arguments prove their own sufficiency, coverage, and existence claims; they are not inferred from Axiom 6 alone.

The following theorem derives the boundary package called *Axiom 6* in downstream applications, with its component warrants and the direction of dependence made explicit. It is a derived theorem here. “Binary, non-eventful, non-parameterized” concerns qualification authority at the unchanged original boundary problem; “non-transmissive” concerns the attempt to obtain its missing authority from structures whose licensed use depends on that problem already being settled.

Theorem 7.5 (Constructed Axiom 6: pre-metric non-transmissivity). *For a pre-fixational proposal interface with independently specified complete retained content, and its determinate evaluated qualification questions, the following boundary constraints hold.*

- (i) *A bare change of representative supplies no additional original-incidence warrant. Where equality exhausts the licensed atom structure, authoritative scalar content is exhausted by equality patterns. At that equality-only interface, an unmarked nonsingleton candidate set admits no lawful equivariant selector choosing an eligible atom. Every invariant metric on the unanchored atom carrier has the discrete equality form, and no invariant strict total order exists when there are at least two atoms. That metric carries only equality/disequality information; tuple-dependent or anchored metric families concern their richer input.*
- (ii) *Qualification is binary at each complete original question. Every separately warranted implementation reports the same original qualification. Neither continuous parameters nor discrete indices can create a further grade or competing value of that qualification at the unchanged locus.*
- (iii) *The qualification relation is non-eventful in the constitutive order: a declared boundary*

operation cannot substitute its own asserted license for the conditions that make the application qualified. In any independently sound grounded rule presentation, an unseeded family of mutually presupposing licenses cannot establish its members.

- (iv) *Additional licensed structure cannot transmit new discriminatory authority backwards through a forgetting map. Its answer descends exactly when it is constant on the retained-data fibres, with coverage required for a total source-level answer. A transfer to a different use requires its own incidence-preservation warrant.*

Thus a claimed source of target-specific authority has three possible accounts: its relevant answer is determined by the retained source data and has an independently established original-incidence warrant; additional fixing data, their license, and their warranted connection to the answer are supplied; or the offered construction has not established its claimed authority. Merely declaring the structure provisional, canonical, latent, metric-free, or post-admissibly licensed does not provide another route.

Proof. For (i), the pre-fixational anti-haecceity argument identifies which changes leave all licensed content unchanged. On the equality-only interface, Theorems 6.3 and 6.7, Proposition 6.5, and Corollary 6.8 give the exact classifications and exceptions. In particular, absence of a selected atom is not used to erase a separately licensed relation.

For (ii), classical qualification has the displayed characteristic value. Original-incidence conservation proves constancy over every family of separately warranted reports of it. No inference from two positive output types to a third truth value is needed or valid.

For (iii), an accountable application must establish its domain and original qualification under Definition 2.3; naming the application does not establish those premises. Replacing them with its asserted eligibility assumes the content whose warrant is required. Theorem 7.4 proves the corresponding exclusion in every grounded rule system satisfying its premises, including infinitary systems. Non-eventfulness concerns this dependency and the fixed relation, not an assertion that no physical process can occur.

Part (iv) is Theorem 7.2 and the explicit target-transfer obligation (7). If a richer answer is not constant on a retained-data fibre, no source-only answer reproduces it. Supplying the missing distinction changes the relevant input and requires its own warrant. If it is constant, it introduces no additional source distinction, although the descended answer must still have its independent semantic warrant. These cases exhaust whether the offered source-only argument or its licensed refinement supplies the requisite determination and warrant. Failure of that argument is not by itself a proof that the physical qualification is false. Provisionality and changes of description do not alter any of the incidence, fixed-point, fibre, or grounding conditions. $\square$

7.6 What downstream papers may invoke

Axiom 6 supports a strong upstream exclusion. A downstream metric, operator, probability, process, or causal model cannot serve as an unexamined source of the very target discrimination and license required to apply that model. It must either be reconstructed from independently warranted source content or enter with its own justified fixing incidence. This applies to symbolic and numerical constructions alike. A discrete index is no exemption, and a dimensionless numerical ratio is not automatically part of a structure-free proposal interface.

Three exact forms of export are especially useful. For an allegedly primitive selector, specify the actual eligible candidates and the transformations preserving all retained authoritative data, then apply the stabilizer test. For a proposed imported quantity or relation, specify the forgetting map and apply the fibre test. For an alleged chain of derivative authorizers,

apply the grounding test: identify the independently grounded judgments and show where the license family first receives a warranted entrance. If it receives none, Theorem 7.4 applies; if it receives one, that source must remain visible in the downstream proof.

The theorem also tells an application what must be preserved in its statement. It must retain the bearer and complete question, the actually licensed structure, the candidate and use coverage it claims, and the original-to-target identification. The phrase “by Axiom 6” cannot itself establish these domain facts. Once they are established, the exclusion follows independently of the chosen presentation, implementation medium, or size of the description space.

This is compatible with all the constructive results that follow. Legitimate fixation can produce an anchor; a complete observation profile can retain warranted distinctions; a licensed operation can construct a new determinate object; a successful new trial can follow a failed one. Their occurrence does not send a new qualification backwards into an unchanged original incidence. The non-transmissive boundary identifies precisely the dependence those constructions must respect.

8 Complete observations and the unique admissibility interface

Symmetry determines which distinctions survive the declared transformation action. An observation interface poses a further question: does it retain every distinction required by its intended uses? Invariance alone does not answer that question. Different orbits may share one observation, and a binary test may omit distinctions visible to other warranted queries. This section proves the exact criterion for sufficiency and the associated uniqueness theorem.

8.1 Observation frames and coverage

Definition 8.1 (Observation frame). An observation frame consists of a set X of candidate presentations, an independently specified comparison equivalence E on X , a set $\mathcal{Q}$ of intended questions, result sets $(V_q)_{q \in \mathcal{Q}}$, and warranted observations

$$P_q : X \longrightarrow V_q.$$

The map $P : X \rightarrow \prod_{q \in \mathcal{Q}} V_q$, $P(x) = (P_q(x))_q$, is its profile. The relation E specifies which presentations are eligible for the declared comparison before equality of their observed answers is tested. It is not assumed to identify every pair with equal profile.

Different coordinate systems may have different raw answer types. Before forming this profile, transport answers to common result types by independently justified identifications, or formulate the comparison in the corresponding typed fibres. A tensor’s raw components are not invariant scalar observations merely because the tensor represents a determinate object. This is the same invariant/equivariant distinction used in Section 4.

An application must explain why $\mathcal{Q}$ covers its claimed uses. A battery report intended only to answer the fixed pass/fail question requires that answer. A report intended also to identify the battery, preserve the original test conditions, state a diagnostic, and authorize specified continuations requires those observations as well. It cannot retain one bit, erase the other work, and call the whole interface complete by definition.

Definition 8.2 (Coverage of a use family). A query family covers a declared family of uses when every result required by those uses is either one of its coordinates or is obtained from those coordinates by an independently sound result construction. Coverage is established from the original use specification, not from the outputs of the proposed reduced interface.

The definition states a proof obligation. It supplies no automatic proof that a finite registry is exhaustive. Coverage can be shown by a syntax induction, a complete physical record specification, a universal property, or a separate semantic classification. The choice of proof depends on the use family. An unrepresented warranted diagnostic defeats an alleged coverage theorem even if it leaves one admission bit unchanged.

8.2 The maximal comparison-respecting quotient

Definition 8.3 (Complete observation equivalence). For an observation frame, define

$$x R y \iff x E y \wedge P(x) = P(y). \quad (8)$$

Write $Q_P = X/R$ and $\pi_P : X \rightarrow Q_P$ for the quotient map.

Theorem 8.4 (Maximal complete-observation quotient). *The relation R is the largest equivalence relation contained in E on whose classes every P_q is constant. Each P_q has a unique descended observation $\bar{P}_q : Q_P \rightarrow V_q$. Moreover, for any set W and map $h : X \rightarrow W$, the following are equivalent:*

- (i) $x R y$ implies $h(x) = h(y)$;
- (ii) there exists a unique $\bar{h} : Q_P \rightarrow W$ with $h = \bar{h} \pi_P$.

Proof. Both E and $\ker P$ are equivalence relations, so their intersection R is an equivalence relation. If $S \subseteq E$ is an equivalence relation preserving all observations, $x S y$ implies both $x E y$ and $P(x) = P(y)$, hence $S \subseteq R$. Each descended observation is defined by $\bar{P}_q([x]) = P_q(x)$ and is well defined by (8).

The same argument defines $\bar{h}([x]) = h(x)$ under (i). Every class has a representative, so π_P is surjective and the factorization is unique. Conversely, a factorization gives equal h -values on every fibre of π_P . $\square$

The result is a maximality theorem: no coarser comparison-respecting equivalence preserves all the observations. It is not a cardinality assertion and does not assume a finite number of distinctions. In universal-property language it is the ordinary quotient construction applied to the complete frame [4]. Its claim to represent every standing-neutral difference depends additionally on the independently established coverage of the standing-bearing uses.

It is useful to retain the comparison component explicitly. Let $\kappa : X \rightarrow X/E$ be the comparison-class map and set

$$\Psi(x) = (\kappa(x), P(x)).$$

Then $\ker \Psi = R$, and Q_P is canonically bijective to $\text{im } \Psi$. This prevents equality of a selected profile from silently identifying presentations outside the licensed comparison relation. It also makes clear why equal observations need not imply a previously unproved physical or mathematical isomorphism.

Theorem 8.5 (Uniqueness over the original presentation space). *Let $\pi_i : X \twoheadrightarrow Z_i$, $i = 1, 2$, be surjections with*

$$\pi_i(x) = \pi_i(y) \iff x R y.$$

There is a unique bijection $u : Z_1 \rightarrow Z_2$ satisfying $u\pi_1 = \pi_2$. It preserves every descended observation. In particular, each Z_i is uniquely isomorphic to Q_P over X .

Proof. Define $u(\pi_1(x)) = \pi_2(x)$. Equal π_1 -values mean xRy , so the definition is well defined. Surjectivity follows from that of π_2 . If $u(\pi_1(x)) = u(\pi_1(y))$, equality of the π_2 -values gives xRy and hence equality of the π_1 -values. The map is therefore bijective. Surjectivity of π_1 makes the commuting equation determine u uniquely. Its preservation of observations follows by evaluating the descended coordinates on an arbitrary $\pi_1(x)$. $\square$

An interface retaining extra harmless metadata need not satisfy $\ker \pi_i = R$; it may distinguish more presentations. Its complete reduction nevertheless does. Thus the uniqueness claim concerns the minimal complete interface in the quotient order, not a mandatory raw encoding or a unique file format.

8.3 The lossless-interface test

The fibre test in Theorem 7.2 concerns determination of one proposed answer from retained data. Here the same principle tests an arbitrary encoding against all required observations and the declared comparison. The lossless-interface test makes the family and its coverage explicit; no new source of qualification authority is introduced.

Theorem 8.6 (Lossless interface criterion). *Let $O : X \rightarrow Z$ be any proposed observation interface and $A : X \rightarrow V$ any required answer. There exists a unique $a : \text{im } O \rightarrow V$ such that $A = aO$ if and only if*

$$O(x) = O(y) \implies A(x) = A(y). \quad (9)$$

For a whole family (P_q) , the interface preserves all answers precisely when $\ker O \subseteq \ker P$. It preserves both all answers and the comparison-class distinction precisely when $\ker O \subseteq R$.

Proof. If $A = aO$, equal interface values have equal answers. Conversely, define $a(O(x)) = A(x)$. Condition (9) proves well-definedness, and the map onto $\text{im } O$ is surjective, proving uniqueness. Apply this result to every coordinate of P and to $\Psi = (\kappa, P)$ for the final assertions. $\square$

No representation grammar occurs in this theorem. It covers every interface map on the specified presentation set. It also supplies a complete falsifier: one pair x, y with $O(x) = O(y)$ but $A(x) \neq A(y)$ disproves sufficiency for that answer. The theorem does not assert that every proposed map satisfies the criterion. Calling O neutral, defining a decoder, or preserving a success bit cannot replace its proof.

Corollary 8.7 (Canonical interface as the least sufficient reduction). *If $\ker O \subseteq R$, there is a unique surjection $d : \text{im } O \rightarrow Q_P$ with $dO = \pi_P$. It is bijective exactly when $\ker O = R$.*

Proof. Use Theorem 8.6 with $A = \pi_P$. Surjectivity follows because every class contains an x . Injectivity is equivalent to xRy implying $O(x) = O(y)$, which supplies the reverse kernel inclusion. $\square$

Example 8.8 (One admission bit and a complete record). Consider valid records from several original battery trials. Two records may both report failure while differing in the battery identifier, original load, or diagnosed interruption. A map retaining only the failure bit is sufficient for the polarity query. It is insufficient for a query whose required answer differs between those records. Both records can be standing-bearing objects; the failure is in the claimed completeness of their reduction, not in their existence. If the intended use requires no diagnostic, omitting it need not fail that narrower task. Coverage is determined by the original task. Appendix A.1 supplies a single explicit incidence model connecting these records, their lawful action, a complete profile, and the inadequate binary reduction.

8.4 Parameters and the impossibility of added co-authority

Proposition 8.9 (Parameter exclusion at a complete interface). *Suppose O satisfies (9) for the original warranted answer A . Let a proposed implementation use extra data θ and report $B(x, \theta)$. If every admitted use of that implementation is accountable for the same determinate original answer at x , then*

$$B(x, \theta) = a(O(x))$$

for every such use. Therefore $O(x) = O(y)$ implies $B(x, \theta) = B(y, \eta)$ whenever the two uses report that same original answer function and have their required warrants.

Proof. By Theorem 2.4, the implementation's original-incidence warrant gives $B(x, \theta) = A(x)$. By Theorem 8.6, $A(x) = a(O(x))$. Repeat at y and use equality of the interface values. $\square$

The parameter may encode useful information, select a permitted computational route, or distinguish representations. It is not required to be literally constant. The conclusion is that it cannot supply discordant standing authority for the unchanged original answer. If varying the parameter changes the claimed answer, either the interface omitted a warranted distinction, the bearer, question, or conditions changed, or the proposed output lacks the original warrant. Identifying the branch requires evidence, not a blanket declaration that all variation is scope change.

This gives a substantive content to “no primitive standing-bearing parameter.” A parameter cannot authorize itself as a replacement for the original incidence. A genuinely warranted additional datum can change the task's available distinctions; it is then itself a determinate construction, necessarily kernel-governed, and its role must be retained in the complete frame. A numerical value and a discrete index are treated by the same argument.

8.5 Uniformity under faithful transport

Theorem 8.10 (Transport of canonical quotients). *Consider two observation frames (X, E, P) and (X', E', P') . Suppose bijections $t : X \rightarrow X'$ and $v : \text{im } P \rightarrow \text{im } P'$ are independently justified and satisfy*

$$xEy \iff t(x)E't(y), \quad P't = vP.$$

Then t induces a unique bijection $\bar{t} : X/R \rightarrow X'/R'$ with

$$\bar{t}\pi_P = \pi_{P'}t.$$

These induced maps preserve identities and composition.

Proof. The displayed hypotheses and injectivity of v give xRy if and only if $t(x)R't(y)$. The formula $\bar{t}([x]) = [t(x)]$ is therefore well defined and bijective, using t^{-1} for its inverse. Surjectivity of π_P gives uniqueness. Identity and composite maps satisfy the required commuting equations; uniqueness identifies them with the induced identity and composite. $\square$

This is the mathematical sense in which the canonical boundary interface is regime-invariant. A faithful change of the whole frame transports its canonical construction. For a noninvertible target update, an induced map still exists if xRy implies $t(x)R't(y)$, but it need not be injective or an equivalence. Such a map does not by itself authorize transport of every old standing claim into the new target. The target incidences require their own warrants.

Neither the quotient theorem nor its transport theorem proves first-order definability, effective extraction, or uniqueness of every physical realization. They establish a semantic universal property over arbitrary sets. That is the generality needed for the boundary; more specialized forms of realization require their own mathematical hypotheses.

8.6 An actual distinction and the data available to a selector

An unobserved distinction can be physically real. Consequently, ignorance of which battery passed must not be used as a proof that the complete physical trial has a symmetry exchanging pass and fail. The following model makes the relation between the incidence theorem and the selection obstruction explicit.

Let $Y = \{a, b\}$ be two candidate identities, and let $X = \{x_a, x_b\}$ be two possible complete configurations. In x_a , the identified passing battery is a ; in x_b , it is b . These configurations represent different original assignments, not two warranted answers to the same fixed assignment. Define

$$A(x_a) = a, \quad A(x_b) = b, \quad J(x, y) \iff y = A(x).$$

Each original incidence is determinate and has a result. The actual trial, not a reader's success, grounds the corresponding answer. Let the available report erase the assignment:

$$O : X \longrightarrow \{*\}, \quad O(x_a) = O(x_b) = *.$$

There is no function $d : \{*\} \rightarrow Y$ with $dO = A$, because its single value would have to be both a and b . This is Theorem 8.6 applied to the exact missing distinction. It excludes even a decoder that is not required to be equivariant, if the decoder claims correctness for both possible original configurations.

Now let a two-element group simultaneously exchange the names a, b and the named configurations x_a, x_b . The original relation is covariant under this action, and A is equivariant. No nonidentity group element fixes either complete configuration. Thus the full data have no stabilizer obstruction to their warranted answer. By contrast, the available report $*$ is fixed, while the two candidate names are exchanged. Its stabilizer has no fixed candidate. The obstruction arises after the information-losing report, exactly where the fibre criterion locates it. This action expresses a renaming of the named assignments; it does not establish a physical process converting the original trial's failure into success.

An anchor repairs the missing identification when it has its own warranted connection to the original assignment. A retained record associating the trial with a supplies a different report at x_a than at x_b . The refined report can then support the answer. Merely appending an arbitrary symbol with no such connection leaves the two cases indistinguishable for the required query. Likewise, a new trial can establish a new performance result; relating it to the original trial remains a separate identification and inference.

This example connects the theorem chain without conflating its domains. The original incidence supplies determinate answers; their simultaneous redescription gives covariance; a reduction must preserve the distinctions needed to recover those answers; the available-data stabilizer can expose a failed reduction. The kernel governs the complete trial, the partial report, the reader, and the later anchor as determinate constructions. Their different information contents do not create optional instances of that governance.

8.7 When symmetry and complete observation give the same quotient

The orbit quotient and the complete-observation quotient answer different questions. The first identifies presentations related by the independently justified action. The second identifies presentations that the declared comparison permits us to compare and whose required answers agree. A theorem is needed before these quotients can be substituted for one another.

Write E_G for the orbit relation of an action of G on X , and retain $R = E \cap \ker P$ from Definition 8.3. All observations in the following statement have already been placed in common scalar result types.

Theorem 8.11 (Exact orbit–observation comparison). *There exists a map $u : X/G \rightarrow X/R$ satisfying*

$$u([x]_G) = [x]_R$$

if and only if $E_G \subseteq E$ and every coordinate P_q is G -invariant. When it exists, u is unique and surjective. It is bijective if and only if, in addition,

$$xEy \wedge P(x) = P(y) \implies y = gx \text{ for some } g \in G. \quad (10)$$

Thus the two canonical quotients coincide over X exactly when lawful orbits are comparison-eligible, the observations are invariant, and the retained comparison and observations separate distinct orbits.

Proof. The proposed formula is well defined exactly when xE_Gy implies xRy . Since $R = E \cap \ker P$, this is equivalent to the two first stated conditions. Every R -class has a representative, so the map is surjective. The orbit projection is surjective and therefore determines u uniquely. Its injectivity says that $[x]_R = [y]_R$ implies $[x]_G = [y]_G$, which is exactly (10). These conditions together give $R = E_G$. $\square$

The separation condition is often the substantial part of an invariant classification. In Proposition 10.1, equality of the full ratio profile is proved to give a common rescaling, so the condition is established by calculation. A single invariant ratio in three coordinates fails it, as the displayed counterexample there shows. Invariance of a proposed observation alone does not establish that it is a complete invariant.

Conversely, an observation family can be complete for its actual task while intentionally distinguishing fewer than all orbits. Let $X = \{u, v\}$, let G be trivial, let $E = X \times X$, and suppose the task asks only a question whose independently warranted answer is 0 at both inputs. Its complete profile is constant. Then X/G has two elements and X/R has one. No required answer has been lost. What fails is the additional claim that this particular task distinguishes every presentation orbit.

There is an opposite caution. If $E = E_G$, even an empty observation family gives $R = E_G$. The retained comparison-class coordinate κ already distinguishes all orbits. This is not a derivation of orbit completeness from no information: the information is carried by the independently specified comparison requirement. Its inclusion must be justified by the original use, such as retaining which trial a report concerns. If no such distinction is required, adding it changes the reduction problem.

Finally, the factorization theorem ranges over all invariant set-functions. A particular physical or mathematical use family may warrant only some of them. The orbit map is a complete invariant as a set-theoretic construction; its formal existence does not prove that a physical reader measures it or that a chosen operational profile reconstructs it. Theorem 8.11 locates that additional proof without making kernel governance optional for either the original object or the reader.

8.8 Exact refinement and why exhaustive coverage matters

Suppose a task is extended by adding questions. The theorem chain should determine exactly what happens to its canonical interface, rather than treating every addition either as harmless notation or as an unexplained new authority.

Theorem 8.12 (Observation refinement). *Let $\mathcal{Q} \subseteq \mathcal{Q}'$ be query families on the same X and comparison equivalence E . Suppose P' restricts to P on $\mathcal{Q}$, and define*

$$R = E \cap \ker P, \quad R' = E \cap \ker P'.$$

Then $R' \subseteq R$ and there is a unique surjection

$$r : X/R' \longrightarrow X/R, \quad r([x]_{R'}) = [x]_R.$$

The map is bijective exactly when every added observation is constant on each R -class, equivalently when it factors through the original complete interface X/R . If one added question distinguishes an R -related pair, the refinement is strict.

Proof. Equality of all P' coordinates implies equality of their restriction, giving $R' \subseteq R$. This inclusion makes the displayed map well defined; representatives prove surjectivity and uniqueness. If every added observation is constant on R -classes, then xRy implies equality of every P' coordinate, hence $xR'y$ and $R = R'$. The converse follows from equality of the relations. Factorization is Theorem 8.4. An R -related pair with unequal added answers belongs to $R \setminus R'$, so the induced surjection is not injective. $\square$

This is an exact information-order statement. A new query can require a finer interface while every previously warranted answer remains conserved. Its additional distinction must be grounded in its original incidence, rather than in a preferred output of the old reduction. A derived query that already factors through X/R enlarges the vocabulary without adding a new distinction. A query that does not factor reveals precisely why the earlier interface was incomplete for the enlarged task.

There need not be a finite subfamily carrying all the information of an infinite profile. For an explicit example let $X = \{0, 1\}^{\mathbb{N}}$, $E = X \times X$, and $P_n(x) = x_n$. The full profile distinguishes every sequence, so R is equality. For any finite $F \subseteq \mathbb{N}$, choose $m \notin F$; the all-zero sequence and the sequence with a single 1 at m agree on every P_n for $n \in F$ but are distinct. Thus no finite subprofile gives the full quotient. Each query is determinate and bivalent, and the full profile is perfectly well defined. Bivalence, finite inspection, and complete profile coverage are different properties.

The example identifies the proof needed to justify a finite sufficient registry in a particular domain: establish that every required coordinate factors through that registry. A family of successful finite tests does not supply that universal statement. Where such a factorization theorem is available, the finite registry has the full warranted force of the complete profile; where it is absent, a stronger claim would outrun the available proof.

9 Constructing and composing a complete interface

An exact interface criterion is useful only if there are methods for proving that a proposed interface satisfies it. This section gives three: descent of partial operations, observation through all declared finite continuations, and assembly by a universal construction. Each starts from independently identified operations and data. None infers generator completeness from the number of displayed outputs.

9.1 Descent of a partial operation

Definition 9.1 (Domain and result compatibility). Let R be an equivalence relation on X , and $f : X \rightarrow X$ a partial operation. It is compatible with R when, for xRy ,

$$x \in \text{dom } f \iff y \in \text{dom } f, \tag{11}$$

$$x \in \text{dom } f \implies f(x)Rf(y). \tag{12}$$

For several input sorts or outputs, use the corresponding equivalences on each sort.

Theorem 9.2 (Exact partial descent criterion). *There is a partial operation $\bar{f} : X/R \rightarrow X/R$ whose definedness and value reproduce f on every representative if and only if (11)–(12) hold. When it exists it is unique, with*

$$\text{dom } \bar{f} = \{[x] : x \in \text{dom } f\}, \quad \bar{f}([x]) = [f(x)].$$

Proof. The domain condition makes membership in $\text{dom } \bar{f}$ independent of the representative. The result condition makes its displayed value independent of the representative. These formulas then give the required reproduction and uniqueness. Conversely, reproduction by one partial function forces equal representatives to have the same definedness and, when defined, the same output class. $\square$

A weak extension that agrees with f where f was defined can still add new defined applications. That is an enlarged partial operation, not an unchanged presentation of the original graph. The theorem does not prohibit constructing the enlarged operation. It identifies exactly which claim of original-operation identity would fail.

Corollary 9.3 (Finite composition). *If f and g are compatible with R , their partial composite $g \circ f$ is compatible with R , and its descended operation is $\bar{g} \circ \bar{f}$. The same holds for every finite word in a family of compatible operations.*

Proof. For xRy , the first application is defined simultaneously at both inputs. If defined, its outputs are R -related. The second application's domain is therefore also entered simultaneously, and its outputs are R -related. This proves both clauses of Definition 9.1. Uniqueness in Theorem 9.2 identifies the composite descent. Induction proves the finite-word statement. $\square$

For operations with several arguments, require compatibility under the product equivalence on all arguments; the same structural induction applies to well-typed terms. This is a construction theorem for a specified language. It does not classify every mathematically conceivable operation or infer finite derivability from semantic determinacy.

9.2 A complete interface for all declared finite continuations

Fix partial updates $f_a : X \rightarrow X$, indexed by a set A , and observations $t_j : X \rightarrow W_j$, indexed by a set T . Adjoin an error state $\perp$ and totalize each update on $X_\perp = X \sqcup \{\perp\}$: an undefined application and every update of $\perp$ return $\perp$. Extend each observation with its own distinct error tag. Include the definedness observation δ with $\delta(x) = 1$ for $x \in X$ and $\delta(\perp) = 0$, even if T is empty.

For a finite word w over A , let f_w apply its letters in their written order; the empty word is the identity. Define

$$x \equiv_{\text{obs}} y \iff t(f_w(x)) = t(f_w(y)) \quad \text{for every finite } w \text{ and every extended } t. \quad (13)$$

The observation profile may be infinite even though each individual continuation word is finite.

Theorem 9.4 (Maximal continuation congruence). *The relation $\equiv_{\text{obs}}$ is the largest equivalence relation on $X_\perp$ preserving the immediate observations and compatible with all totalized updates. It induces a definedness-preserving quotient of the original partial system on X .*

Proof. Equality of the displayed profile is an equivalence relation. The empty word shows preservation of immediate observations. If $x \equiv_{\text{obs}} y$, then for any letter a , word w , and observation t ,

$$t(f_w(f_a(x))) = t(f_{aw}(x)) = t(f_{aw}(y)) = t(f_w(f_a(y))).$$

Thus $f_a(x) \equiv_{\text{obs}} f_a(y)$. Conversely, any observation-preserving equivalence compatible with every update remains compatible along each finite word by induction, and is consequently contained in (13). Finally, δ separates $\perp$ from every ordinary state. Related ordinary states therefore cannot be sent to $\perp$ on one side and an ordinary state on the other. This supplies the original partial domain and result conditions. $\square$

The proof is a behavioral quotient argument of the kind studied in automata and coalgebra [5]. Here its universe of future uses is explicit. It is complete for the specified finite words and observations. If a target additionally observes a limit process or another operation, that use needs a coordinate or a separate proof that it is determined by this profile. Likewise, a finite word containing a noncomputable primitive is a finite expression, not an effective algorithm.

Example 9.5 (Retesting after a record reduction). Suppose two original records have the same pass/fail bit but produce different warranted continuations under the declared test-use protocol. The one-bit reduction is not a congruence for that protocol. Including all admitted continuation observations separates the records exactly where the protocol can distinguish them. If a retest changes the battery's conditions, the target and transition semantics must record that change rather than describe it as a new answer about the unchanged first trial.

9.3 Assembly from components and overlaps

Assembly provides another construction method, starting from a small diagram $D : \mathcal{I} \rightarrow \mathcal{C}$. Objects represent independently warranted component interfaces; arrows represent their warranted overlap or incidence maps. The category $\mathcal{C}$ must specify lawful interfaces and lawful morphisms. Establishing that a proposed assembly is a colimit *in this category* is a substantive obligation. An ordinary set construction alone does not establish semantic lawfulness.

Theorem 9.6 (Uniqueness of a warranted universal assembly). *If $(L, (\lambda_i))$ and $(L', (\lambda'_i))$ are colimit cocones of the same independently fixed diagram in $\mathcal{C}$, there is a unique isomorphism $u : L \rightarrow L'$ with $u\lambda_i = \lambda'_i$ for every i .*

Proof. The universal properties give unique morphisms $u : L \rightarrow L'$ and $v : L' \rightarrow L$ commuting with the respective cocones. Both vu and id_L commute with every λ_i , hence are equal by uniqueness of the mediator out of L . Similarly $uv = \text{id}_{L'}$. The cocone equation determines u uniquely. $\square$

For a set diagram, existence can be proved directly: take the disjoint union of all component sets and quotient by the equivalence relation generated by $x \sim D(a)(x)$ for every diagram arrow a . A compatible family of maps descends uniquely because it has equal values on these generating identifications. If the relevant interfaces carry additional structure or incidence constraints, prove that this construction retains them and has the universal property for their lawful morphisms. This is the actual assembly burden.

Proposition 9.7 (Induced observations after assembly). *Given a colimit cocone in $\mathcal{C}$ and a compatible family of lawful observation maps $p_i : D(i) \rightarrow V$, there is a unique lawful map $p : L \rightarrow V$ with $p\lambda_i = p_i$. Completeness of p for a claimed use family additionally requires that every such use be determined by these component and incidence observations.*

Proof. Existence and uniqueness of p are the colimit universal property. That property asserts how compatible maps assemble, not that a chosen family of maps observes every intended distinction. The latter statement is precisely the separate coverage obligation of Definition 8.2. $\square$

The dependency order is therefore source data, diagram, proved assembly, induced observations, coverage, and complete reduction. Selecting the diagram or its probes after inspecting which candidates one wishes to exclude reverses this order. The concrete certificate-fibre development follows this constructive pattern [15]; its physics-specific existence and coverage statements are not imported here.

9.4 Compatibility and realization

Even complete local interfaces require joint compatibility. If $C_i \subseteq \Omega$ is the independently defined set of realizations allowed by component i on a common target, a simultaneous realization belongs to $\bigcap_i C_i$. Pairwise nonempty intersections do not ensure this set is nonempty: $\{1, 2\}$, $\{2, 3\}$, and $\{1, 3\}$ give a finite counterexample. Shared names or pairwise agreements are weaker still.

A role assignment similarly requires an independently specified candidate set, eligible occupants, the complete compatibility relations, and a lawful transformation action. Only after existence is proved can one claim that the compatible assignment space has exactly one orbit. A single orbit can contain many raw assignments; an empty space has no realization. Neither role counts nor a binary admission label establishes either existence or orbit uniqueness.

These observations locate the boundary's downstream work. It requires the realization proof to retain its target, witnesses, and comparisons. It does not replace that proof with a universal assertion that all possible source data yield the same physical object.

10 Realization and the warrant for selection

These worked examples show how independently warranted fixing data or an intrinsic mathematical selection criterion supplies a particular answer under the preceding boundary constraints. They distinguish three achievements: identifying a class of equivalent presentations, selecting an element of a candidate class, and warranting that selected element as a physical realization. Each achievement has its own proof obligations. Necessary kernel governance applies to every determinate object involved; the examples identify what further mathematical structure supplies a particular answer. Neither example classifies all physical theories.

10.1 A complete ratio profile and its unresolved scale

Fix $n \geq 2$ and let $X = (\mathbb{R}_{>0})^n$. The coordinate labels are retained: x_i and x_j have independently specified roles. The positive scalars act by common rescaling,

$$c \cdot (x_1, \dots, x_n) = (cx_1, \dots, cx_n), \quad c > 0.$$

This action models a comparison that retains relative magnitudes and forgets common scale. Its physical interpretation must be supplied separately. Rescaling may describe a change of units or a change between physically distinct states; the orbit calculation alone does not decide which interpretation applies.

Define the ratio profile

$$\rho(x) = \left(\frac{x_1}{x_n}, \dots, \frac{x_{n-1}}{x_n} \right) \in (\mathbb{R}_{>0})^{n-1}. \quad (14)$$

Positivity ensures that every displayed ratio is defined. The distinguished index n is an explicit coordinate choice for presenting the quotient. Choosing another denominator produces an equivalent ratio description and introduces no absolute scale.

Proposition 10.1 (Complete invariant for common rescaling). *For $x, y \in X$, one has $\rho(x) = \rho(y)$ if and only if $y = cx$ for a unique $c > 0$. Consequently every function $F : X \rightarrow Y$ satisfying $F(cx) = F(x)$ factors uniquely through ρ .*

Proof. Common rescaling cancels in each ratio. Conversely, if the profiles agree, set $c = y_n/x_n$. For $i < n$, ratio equality gives $y_i = cx_i$, and the same equation holds for $i = n$ by the definition of c . Positivity of x_n makes c unique. For factorization, define

$$\bar{F}(r_1, \dots, r_{n-1}) = F(r_1, \dots, r_{n-1}, 1).$$

Since $x = x_n(\rho(x), 1)$, invariance gives $F = \bar{F} \circ \rho$. Surjectivity of ρ gives uniqueness. $\square$

The factorization statement concerns outputs whose meaning is unchanged by rescaling. A covariant quantity has a different transformation law. For example, $M(x) = x_1 + \dots + x_n$ satisfies $M(cx) = cM(x)$, so it cannot factor through the ratio profile as an absolute scalar. That failure is expected: its output still carries scale. Retaining a covariance law is mathematically consistent without fixing a numerical value. The profile is complete for the declared invariant questions, not for a larger family that also asks absolute magnitudes. Enlarging the query family requires retaining the corresponding scale information.

Thus the ratios carry the entire information invariant under this action. They do not fix x_n : the fibre over r is exactly

$$\rho^{-1}(r) = \{a(r_1, \dots, r_{n-1}, 1) : a > 0\}. \quad (15)$$

Completeness matters here. For three coordinates, the single ratio x_1/x_2 is invariant but incomplete: $(1, 1, 1)$ and $(1, 1, 2)$ agree on that ratio and belong to different common-scale orbits. A formula that happens to be invariant therefore need not carry every invariant distinction. Proposition 10.1 justifies the full profile by proving both directions of the orbit criterion. This is the application-specific completeness argument required before a reduced record can replace the original data for all scale-invariant questions.

The action is free, and this fibre contains one point for every positive scale. The section $r \mapsto (r, 1)$ is a legitimate normalization. Its existence shows why an undifferentiated ban on canonical representatives would be mistaken. It chooses a convenient presentation of the ratio class; it does not show that the last physical quantity has value one in an independently fixed unit.

Proposition 10.2 (Reconstruction from one anchor). *The map*

$$X \longrightarrow (\mathbb{R}_{>0})^{n-1} \times \mathbb{R}_{>0}, \quad x \longmapsto (\rho(x), x_n)$$

is a bijection, with inverse $(r, a) \mapsto a(r, 1)$.

Proof. The displayed maps are inverse by direct substitution. $\square$

For example, the profile $(2, 3)$ fixes the ray $\{a(2, 3, 1) : a > 0\}$. An independently warranted anchor $x_3 = 5$ reconstructs $(10, 15, 5)$. The reconstruction is exact relative to those inputs, but its warrant includes the source of the anchor. If the anchor is measured or fitted, the resulting absolute values retain that dependence. If it is independently derived, that derivation supplies the missing scale premise. An arbitrary choice of a does neither. The same accounting applies to the ratios themselves: the algebraic reconstruction does not establish how their numerical values were obtained.

A partially specified anchor yields a correspondingly partial determination. If its independently established content is only $a \in [a_-, a_+]$ with $0 < a_- \leq a_+$, the compatible vectors

form exactly $\{a(r, 1) : a \in [a_-, a_+]\}$. Their coordinates remain linked by one shared scale; independent coordinate intervals would forget that relation. Selecting the midpoint supplies a reporting convention, not a proof that the physical anchor has that value. Conversely, a singleton anchor collapses the compatible set without introducing a further selector. The shape of the reconstructed output thus records precisely how much of the scale question the anchor answers.

The distinction is between the output types already present in the calculation: a ratio class, a normalized representative, and an anchored vector. Promoting one to another requires the corresponding datum and its justification. No probability on the scale fibre, ranking of its members, or preference for a numerically simple anchor is supplied by the quotient theorem.

10.2 Self-adjoint extensions and the symmetry constraint

Let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be a closed, densely defined symmetric operator on a complex Hilbert space. Its domain is part of the operator, not an omitted detail. Put

$$N_+ = \ker(T^* - iI), \quad N_- = \ker(T^* + iI),$$

and assume $0 < \dim N_+ = \dim N_-$. The common dimension may be infinite. Von Neumann's parameterization associates self-adjoint extensions with the unitaries $U : N_+ \rightarrow N_-$:

$$D(T_U) = D(T) \dot{+} \{\eta + U\eta : \eta \in N_+\}, \quad T_U = T^*|_{D(T_U)}. \quad (16)$$

The correspondence is one-to-one [9].

The domain issue is genuinely infinite dimensional. In a finite-dimensional Hilbert space a dense linear domain is the whole space, and a symmetric operator is already self-adjoint. Its deficiency spaces are zero, so the present nonzero-deficiency hypothesis does not apply. For an unbounded operator, by contrast, a formal expression does not determine its admissible domain. The extension problem retains the expression and its original applications while asking which larger domain supports a self-adjoint realization. This is why the domain must remain visible in the comparison.

Consider complex-linear unitary redescriptions of the original operator base,

$$G_T = \{W \in \mathcal{U}(\mathcal{H}) : WD(T) = D(T), WT\psi = TW\psi \text{ for all } \psi \in D(T)\}.$$

Additional retained structure may restrict the permitted group to a subgroup $G \subseteq G_T$. The statements below apply to either choice. Antiunitary transformations are not included in this action.

Proposition 10.3 (Invariant extension parameters). *For $W \in G$, write $W_\pm = W|_{N_\pm}$. The action induced on extension parameters is*

$$U \longmapsto W_- U W_+^{-1}. \quad (17)$$

Hence T_U is G -invariant precisely when

$$W_- U = U W_+ \quad (W \in G). \quad (18)$$

The invariant parameters are exactly the unitary intertwiners between the two deficiency representations. Their set is either empty or contains a circle of distinct extension parameters through each of its elements.

Proof. From $WTW^{-1} = T$, taking adjoints gives $WT^*W^{-1} = T^*$. Thus W preserves both deficiency spaces. Applying W to the domain in (16) and substituting $\zeta = W_+\eta$ gives

$$WD(T_U) = D(T) \dot{+} \{\zeta + W_-UW_+^{-1}\zeta : \zeta \in N_+\}.$$

The action formula and fixed-point criterion follow. This is the usual intertwiner criterion for invariant extensions [9, Theorem 3.5]. If U satisfies (18), then so does $e^{i\theta}U$ for every $\theta \in \mathbb{R}$. Since $N_+ \neq \{0\}$, distinct phases give distinct unitaries and therefore, by the one-to-one parameterization, distinct self-adjoint extensions. $\square$

Invariance alone consequently does not single out one member of the full extension family under these hypotheses. This is a statement about the solution set of (18). It does not exclude an intrinsic rule using additional properties of the fixed operator. Nor does equal deficiency dimension alone establish that the empty alternative occurs: equality of dimensions and unitary equivalence of the two group representations are different conditions. When further hypotheses guarantee an invariant extension, emptiness is excluded. Semiboundedness provides an important example below.

The original graph also fixes a precise preservation claim. Every T_U agrees with T on $D(T)$, but a proper extension supplies values on a larger domain. Such an extension is a new operator realization with the original operator as a restriction. Its legitimacy does not make its new applications applications of the unchanged original graph. For a fixed $\psi \in D(T)$, every extension answers the original value query by $T\psi$. For a vector outside $D(T)$, whether evaluation is licensed and what it returns depend on the selected extension. The complete question now includes that extension's domain and parameter. Two answers obtained from different T_U therefore need not be discordant answers about one fully specified operator. This distinction permits extension theory while preserving the identity of the operator being extended.

10.3 An intrinsic selection criterion: the Friedrichs extension

Suppose additionally that $T \geq mI$ for some $m \in \mathbb{R}$. Its nonnegative form

$$q_T[\psi] = \langle \psi, (T - mI)\psi \rangle, \quad \psi \in D(T),$$

is closable. The unique operator representing its closure, shifted back by mI , is the Friedrichs extension T_F [11, Chapter VI]. This construction supplies an exact selection rule from the given operator and Hilbert-space structure. It requires no arbitrary deficiency-space phase.

To state its extremal content, let

$$\mathcal{E}_m(T) = \{A = A^* : T \subseteq A, A \geq mI\},$$

and let q_A be the closed nonnegative form of $A - mI$. The closed-form order is

$$\begin{aligned} A \preceq_q B &\iff D(q_B) \subseteq D(q_A), \\ &\quad q_A[\psi] \leq q_B[\psi] \quad (\psi \in D(q_B)). \end{aligned} \tag{19}$$

A larger element has a smaller form domain and a greater or equal form value on that domain. This order compares closed forms; it is not inclusion of the operator graphs.

Proposition 10.4 (Friedrichs endpoint). *For $B \in \mathcal{E}_m(T)$,*

$$B = T_F \iff A \preceq_q B \text{ for every } A \in \mathcal{E}_m(T).$$

The assignment $T \mapsto T_F$ commutes with unitary conjugacy.

Proof. For every $A \in \mathcal{E}_m(T)$, the closed form q_A agrees with q_T on $D(T)$. It therefore extends the closure of q_T . Hence $D(q_{T_F}) \subseteq D(q_A)$ and the forms agree on $D(q_{T_F})$, giving $A \preceq_q T_F$. This is the classical greatest-extension property [10]. If B is also greatest, mutual comparison gives equal form domains and equal quadratic forms. Polarization and uniqueness of the represented operator yield $B = T_F$. Finally, unitary conjugacy carries q_T and its form closure to the corresponding forms for WTW^{-1} . Uniqueness of the represented operator gives

$$(WTW^{-1})_F = WT_FW^{-1}.$$

□

In particular, T_F is invariant under every element of G_T . Thus the empty intertwiner alternative is unavailable in this semibounded setting. The phase family from Proposition 10.3 remains a family of invariant self-adjoint extensions, but its members need not all retain the lower bound m or satisfy the Friedrichs endpoint condition.

The selection warrant is now visible: close the original semibounded form, or equivalently select the greatest member of the specified form order. This is substantive analytic structure beyond the invariance test. It can be intrinsic to the fixed data, so the example supplies no general prohibition on canonical extensions. Adopting this mathematical realization for a particular physical system still requires justification of the operator, domain, and realization criterion appropriate to that system. The ratio example and the operator example therefore exhibit the same discipline: a complete invariant determines what survives the declared comparison, while any stronger selected output must be supported by the additional information or criterion that actually fixes it.

11 Historical fixation and the exact endpoint condition

Irreversibility in the kernel preserves the original occurrence and its conditions. A dynamic model can impose stronger irreversibility, but that additional claim must be proved for its transitions. The following results distinguish these objects exactly.

Let X be a set of fully specified stage objects, let $\rightarrow$ be a legal transition relation, and let $S : X \rightarrow \{0, 1\}$ express the relevant standing qualification. A path is a nonempty finite sequence $h = (x_0, \dots, x_n)$ with $x_i \rightarrow x_{i+1}$ for all applicable i . Define

$$C(h) \iff S(x_i) = 1 \text{ for every } i \leq n.$$

The relation and predicate are fixed before deciding whether a path is coherent. Later knowledge of their values is not a change of the underlying path.

Theorem 11.1 (Retained-history conservation). *Every prefix of a coherent path is coherent. If a path is incoherent, every extension retaining it as a prefix is incoherent. For composable legal paths h, k , with their common endpoint identified,*

$$C(h \cdot k) \iff C(h) \wedge C(k).$$

Proof. Coherence is the conjunction of the standing assertions for all retained stages. A prefix removes conjuncts; extension retains every original conjunct. A false original conjunct therefore remains a witness to incoherence. The stages of a concatenation are exactly those of the constituent paths, with a common endpoint possibly listed once instead of twice; this gives the displayed equivalence. □

The historical result needs no assumption that a later stage is negative. It applies when the battery is recharged and a later trial succeeds: the larger history contains both the original failed trial and the successful new one. Calling that history wholly successful would delete a retained fact. Calling its latest trial successful can be correct. These are different questions.

Theorem 11.2 (Endpoint certification if and only if negative absorption). *The following are equivalent:*

- (i) *Every legal edge preserves negativity: $x \rightarrow y$ and $S(x) = 0$ imply $S(y) = 0$.*
- (ii) *Every finite legal continuation preserves negativity.*
- (iii) *For every legal nonempty finite path h ,*

$$C(h) \iff S(\text{last}(h)) = 1.$$

Proof. Condition (i) iterates along a finite path, yielding (ii), including the length-zero case. Under (ii), a positive endpoint cannot have a negative earlier stage, so the right side of (iii) implies $C(h)$. Its converse follows from the definition of C . Finally, suppose (iii) and take a legal edge $x \rightarrow y$ with $S(x) = 0$. If $S(y) = 1$, the two-stage path would be coherent by (iii), contradicting its first stage. Thus (i) holds. $\square$

This theorem states the exact burden of the endpoint shortcut. The existence of one non-returnable event does not prove condition (i). Neither does the historical fact that a failed original use remains a failure of that original use. A transition relation allowing a newly charged battery trial to replace a failed trial at the level of current-state qualification need not be negative-absorbing.

Example 11.3 (A canonical history-state realization). Let the new state space consist of the legal histories themselves. A transition appends one legal stage, retaining the entire earlier history. Give a history standing exactly when $C(h)$ holds. By Theorem 11.1, this new state predicate is negative-absorbing, so Theorem 11.2 applies to paths of history-states. A strict append also increases the history's length and cannot return to the same complete history under further appending. The projection taking a history to its last physical stage may return to a previous stage and may discard an earlier failure. Thus non-return and absorption are realized for the explicitly constructed history object, without being imposed on every projected physical state.

The distinction is structural rather than merely verbal. A photographic record of a trial and the trial's final arrangement need not retain the same information. A reversible equation for an arrangement does not invert the record's full ancestry. Conversely, the historical construction does not establish a thermodynamic arrow or a rate of entropy production. It preserves what has been fixed, with the fixed object and its projection kept distinct.

11.1 The order induced by retained consequences

Proposition 11.4 (Historical order reflection). *Let $\preceq$ be an independently established preorder of retained historical consequence. Define $x \sim y$ when $x \preceq y$ and $y \preceq x$. Then*

$$[x] \leq [y] \iff x \preceq y$$

is a well-defined partial order. Every monotone map from the preorder to a partially ordered set factors uniquely through this quotient by a monotone map.

Proof. Reflexivity and transitivity make $\sim$ an equivalence relation. Replacing x, y by mutually preceding representatives preserves $x \preceq y$ by transitivity, establishing well-definedness. Reflexivity and transitivity pass to the quotient; two classes ordered in both directions are equal by definition of $\sim$. If f is monotone into a poset, $x \sim y$ gives $f(x) \leq f(y)$ and $f(y) \leq f(x)$, hence $f(x) = f(y)$. The usual quotient construction gives the unique factor, and its monotonicity follows from that of f . $\square$

For example, fix a complete historical consequence assignment $H(x)$ and put $x \preceq y$ exactly when $H(x) \subseteq H(y)$. The quotient is the image of H ordered by inclusion. A strict enlargement prohibits return along continuations that preserve this inclusion. To apply that statement to actual updates, prove their monotonicity for the original consequence assignment. A list of words such as “retains,” “transports,” and “presupposes” does not establish that the corresponding relations compose.

The retained-path proofs are elementary and coincide with the independently reusable historical portion of the trajectory development [16]. The stronger endpoint equivalence here identifies its precise additional hypothesis. The resulting package retains historical irreversibility, admits accurate diagnosis and new performances, and prevents an endpoint shortcut from quietly becoming a universal law of physical dynamics.

12 The universal boundary theorem and its interpretation

The preceding results can now be assembled without placing observational agreement in the definition of the comparison class. The original bearer, question, transformation action, and use family are supplied first. Individual warrants, invariant factorization, and interface completeness then perform separate jobs.

Theorem 12.1 (AMetric boundary: necessity, exact discrimination, and canonical conservation). *Consider a non-degenerate determinate construction, its independently specified result incidences, and a complete warranted observation frame for a declared family of uses. For its independently justified lawful presentation action and comparison structure:*

- (i) *Reference, standing, admissibility, and irreversible fixation are necessary functional roles. Every accountable presentation of the same determinate incidence gives the same answer.*
- (ii) *Pre-fixational authority obeys the anti-haecceity and non-transmission constraints of Theorem 7.5. Equality-only classification cannot acquire an unlicensed atom discriminator. A richer answer descends to retained source data exactly under Theorem 7.2; an unseeded family of derivative licenses cannot establish itself in a grounded rule system satisfying Theorem 7.4.*
- (iii) *Every invariant scalar observation factors uniquely through the presentation-orbit quotient. Equivariant selection is possible exactly through eligible stabilizer-fixed outputs on the input orbits, with the orbitwise choice data of Theorem 4.6 supplied. It is unique exactly when the eligible fixed-point set at every input is a singleton, without an axiom of choice.*
- (iv) *Covariance of an independently specified functional result relation is equivalent to invariance of its exact domain and equivariance of its result function. For reversible comparisons between different output types, coherent selection has the component and loop-fixed-point classifications of Theorems 5.6 and 5.7.*
- (v) *On an equality-only finite-tuple interface, scalar observations and atom-valued equivariant selections are exactly those classified in Theorems 6.3 and 6.7. Further discrimination requires independently warranted additional data of the kind tested in Theorem 4.9.*

- (vi) *The complete observation frame has a unique maximal comparison-respecting quotient. A proposed interface preserves a required answer exactly when it satisfies Theorem 8.6. Minimal complete interfaces are uniquely isomorphic over the original presentations. The orbit and observation quotients coincide exactly under Theorem 8.11; added queries refine the quotient exactly as in Theorem 8.12.*
- (vii) *The canonical observation quotient commutes with faithful transport of the complete frame. Partial operations descend under the stated domain and result conditions; warranted universal assemblies are unique up to the cocone-preserving isomorphism.*
- (viii) *Retaining an original history preserves its original successes and failures. Endpoint standing certifies every stage of every finite legal path exactly under the absorption condition of Theorem 11.2.*

Consequently a change of presentation supplies no additional co-authority for a discordant warranted result at an unchanged determinate locus. This conclusion is uniform over representation types, languages, implementation media, and the cardinalities of the observation and candidate sets.

Proof. Part (i) is Proposition 2.2 and Theorem 2.4, extended pointwise by Theorem 2.6. Part (ii) is the constructed boundary theorem and its stated fibre and grounding results. Part (iii) is Theorems 4.2, 4.6, and 5.4. Part (iv) is Theorem 5.2 and the two stated groupoid theorems. Part (v) is the equality-pattern and stabilizer classification and the anchor theorem. Part (vi) follows from Theorems 8.4, 8.6, 8.5, 8.11, and 8.12. Part (vii) follows from Theorem 8.10, Theorem 9.2, and Theorem 9.6. Part (viii) is Theorems 11.1 and 11.2. The consequence follows directly from Corollary 2.5 and Proposition 8.9. None of those arguments enumerates the possible implementations or requires a finite complete query family. $\square$

12.1 The dependency chain and the force of the universal claim

Theorem 12.1 assembles the seven commitments mapped in §1.1. The constitutive argument supplies necessary governance; independently warranted incidences supply conservation; the fibre, stabilizer, lossless-interface, and grounding tests expose the corresponding mathematical obligations. Theorem 7.5 is the central transition from pre-fixational classification to the non-transmissive boundary. The interface and construction results retain their own proofs and hypotheses.

The assembled result has three distinct universal quantifiers. For *any* non-degenerate determinate construction, the constitutive dependence holds. For *any* separately warranted presentations of a fixed determinate result, conservation holds. For *any* sets, actions, families, and interfaces satisfying the displayed mathematical hypotheses, the corresponding exact theorems hold. None of these quantifiers ranges only over representations named in a finite grammar or records listed by a particular researcher. Nor do they assert that every arbitrary map has the properties demanded of a standing-bearing use.

12.1.1 Why a short consequence can carry a substantive restriction

The equality step in same-incidence conservation is elementary once single-valuedness and the two incidence warrants are established. Its content is not a new proof that functions are single-valued. Its role is to locate every legitimate source of authority before comparison of rival outputs. A different notation, reader, or implementation cannot add a second warranted value at that unchanged locus. The domain may nevertheless contain more results, different

questions, and new constructions. Their warrants must be supplied where their own incidences enter the chain.

The same distinction matters for the quotient and selector results. Their general mathematics is established mathematics. The substantive application is the identification of the actual comparison, original candidate conditions, sound observations, and required use family. The exact theorems then make the consequences of those facts explicit and testable. The operator example illustrates this division particularly clearly: symmetry supplies an intertwiner condition, while semibounded form closure supplies a stronger intrinsic selection criterion. The latter earns a uniqueness conclusion that the former alone does not supply.

The argument would be circular if agreement were used to identify the original relation, if covariance were asserted only because a desired function happened to be equivariant, or if completeness were obtained by deleting inconvenient queries. It is not circular to establish original data independently and then derive agreement or a canonical interface from them. The independent data can come from a definition of a mathematical object, a domain theorem, or an actual physical incidence. What matters is that the premise be warranted for its own content, rather than merely restating the rival-output conclusion being sought.

12.1.2 What a universal necessity theorem contributes downstream

A downstream theory must supply its actual entities, equations, identification maps, and realization arguments. Those constructions incur the kernel by being determinate, rather than by accepting an additional authorization rule. The AMetric results constrain the claims that can then be made: whether a proposed redescription preserves an original answer, whether a selection has acquired the needed discrimination, whether a reduced interface retains every required use, and whether an endpoint preserves what is claimed of its history.

If each input's eligible stabilizer-fixed set is a singleton, the corresponding selector is unique. If complete invariant data separate candidate orbits, then fixing a jointly realized profile and comparison class determines exactly one candidate orbit. A proved lawful universal property gives a unique assembly up to its source-preserving isomorphism. These are positive routes to stronger downstream theorems. Their independent existence, coverage, and identification proofs are part of the construction's content. Keeping them explicit is how the theorem chain supports the strongest justified conclusion without allowing a preferred conclusion to supply its own missing warrant.

12.2 Why the constraint is necessary for mathematics and physics

In mathematics, a result, derivation, operator application, or endpoint is offered as the same object across expressions and arguments. The claim must retain its conditions of identity and valid use. In physics, a state, carrier, record, or experimental endpoint likewise supplies actual relata and conditions; a description cannot replace them while retaining the claim to describe that incidence. The kernel identifies the common necessary work. The boundary identifies the discrimination and transport that this work can support.

No independently complete physical object is first established and then optionally decorated with the four roles. Its determinate incidence already performs their work. Equally, the kernel does not replace a physical existence theorem, a measurement relation, a field equation, or a boundary condition with four words. Those substantive constructions establish the relevant incidences. Their faithful comparison then necessarily obeys the boundary.

This makes the result an upstream constraint on downstream mathematics and physics. It tells a realization proof what it must preserve and tells a selection proof what independent

fixing work it must supply. It does not infer a specific theory's unique realization from bivalence alone. Distinct source data can support distinct determinate objects while every one incurs the same constraint.

12.3 Uniqueness without an assumed answer

The uniqueness argument separates its independently warranted premises from the agreement derived from them. Result determinacy is established for the original incidence relation; it is not obtained by choosing the outputs that agree. Query coverage comes from the intended uses; it is not obtained by deleting tests on which two encodings differ. The lawful action is established from transport of the data; it is not chosen to preserve a desired verdict. A colimit is proved in the relevant category; it is not merely given a universal name.

Once those burdens are met, the uniqueness results are exact. All warranted answers to the unchanged determinate question agree; if warranted existence is established, there is exactly one. There is one largest comparison-respecting observation equivalence, and one quotient interface up to the unique comparison commuting with the original presentation maps. Where independently warranted components admit a universal assembly, its uniqueness is an additional construction theorem. The shortness of the final factorization proofs reflects the precision of these burdens, not their disappearance.

The result does not claim new group theory or a new set-theoretic quotient principle. Its contribution is an explicit necessity-to-interface argument that exposes what a standing-bearing use must preserve, provides complete tests for selection and reduction, and distinguishes those tests from their domain-specific realization. That distinction is what allows an alleged extra authority to be evaluated without building its exclusion into a comparison definition.

12.4 The remaining live tests

The main claims expose concrete defeat routes.

1. *Constitutive entailment.* Challenge an inference from actual determinate objecthood to one of the four functional commitments in Proposition 2.2, identifying the commitment that is claimed not to follow and the determinate construction alleged to sustain its absence. Alternatively, show that a reconstruction incorrectly attributes a role to a mechanism. Where the mechanism does perform the required work, its different name or distribution is an implementation, not an omission.
2. *Incident discordance.* Establish different original-incidence results at the same bearer and complete query where determinacy was claimed. This challenges the actual relation or its determinacy proof.
3. *Information loss: fibre and lossless-interface tests.* Exhibit equal retained interface values with different required warranted answers, or an intended use not covered by the proposed profile.
4. *Incorrect symmetry: stabilizer test.* Exhibit relevant authoritative structure not preserved by a transformation included in the claimed lawful action, or an alleged eligible selector failing its stabilizer test.
5. *Unsupported license: grounding test.* Exhibit a proposed license whose claimed derivation has no independent ground, or identify an unsound rule or an incorrectly declared dependency. Theorem 7.4 gives the exact exclusion when every entrance into an unseeded

family already requires a family member. A genuine grounded entrance defeats that no-entry premise; it does not by itself prove every license in the family.

6. *Failed construction or export.* Exhibit an omitted candidate, a missing domain case, incompatible required incidences, or a failed universal property. A target change must be identified by an independently changed coordinate; it cannot be inferred merely from disagreement with a preferred result.

These tests are semantic. A finite registry can make them easier to inspect but cannot close an unproved coverage claim by displaying no recorded counterexamples. Nor does failure to find a computation establish failure of the underlying determinate incidence. The implementation distinctions are stated in the appendices.

12.5 Conclusion

The AMetric boundary is the invariant constraint on what can acquire standing through a determinate construction. Its pre-metric foundation excludes unlicensed representative privilege, derives the equality-only limits on discrimination, and prevents post-fixational authority from supplying its own missing ground. Constructed Axiom 6 states that dependency together with binary qualification and the exact backward-determination test. Its general form is exact: invariant information descends to orbits; eligible selection requires stabilizer-fixed outputs; complete warranted observations determine a canonical maximal interface; and the same determinate incidence cannot sustain discordant co-authoritative answers. Additional fixation is possible, but its discriminating contribution requires its own warrant and remains subject to the same kernel.

The equality-only case, metric anchoring, operator endpoint selection, and retained-history construction show both sides of the boundary. They identify what is unavailable before the requisite data are fixed and what becomes available once the missing work is supplied. The resulting universal constraint does not depend on an optional interpretive overlay, a finite catalogue of representations, or an assumption that all legitimate construction is computable. Its application is as strong as the independently established incidence, coverage, and realization arguments it governs.

A Finite certificates and semantic conservation

The semantic theorems quantify over arbitrary presentation and query sets. A finite implementation is an additional construction. The following result states a sufficient proof-preservation criterion without inferring effective finite derivability from the mere existence of a determinate object.

Definition A.1 (Locally compilable certificate system). Fix an original judgment system $\mathcal{J}$ and its derivations. A finite certificate is a finite directed acyclic graph whose nodes carry judgments and are either:

- (i) leaves with supplied original derivations, including any explicitly licensed zero-premise rule; or
- (ii) applications of declared constructors to the judgments of their predecessor nodes, carrying the actual original domain and side-condition warrants required by that application.

For every constructor, a local translation takes original derivations of its premises, together with its original domain and side-condition warrants, to an original derivation of its conclusion. If an

effective compiler is claimed, the graph, leaf derivations, domain and side-condition warrants, and every such translation are supplied in effective representations, with the translations effective on those representations.

Theorem A.2 (Finite certificate compilation). *Every certificate satisfying Definition A.1 supplies an original derivation of each node's judgment, in particular its endpoint. Under the effectiveness clause, those derivations can be computed from the supplied certificate data.*

Proof. A finite acyclic graph has a topological ordering. At a leaf, use the supplied original derivation. At any later node, all predecessor derivations have already been obtained; apply the declared local translation with its side-condition warrants. Induction over the finite ordering constructs all required derivations. Under the effectiveness clause, a topological ordering can be computed from the finite graph and each translation can be applied to the supplied effective data, so the same induction is an algorithm. $\square$

The conclusion concerns the stated certificate extension. It does not eliminate an arbitrary infinitary rule. For such a rule, one would additionally need a theorem that every accepted application provides a certificate of this form and that the local translation preserves its intended judgment. A finite output string is not such a theorem. A finite observation expression may also call a noncomputable test.

Likewise, absence of a supplied certificate is not a proof of semantic non-incidence. In the battery example, an unread original record can support a true result that a particular reader has not yet established. A claimed use requiring a completed verification must instead ask the different, explicit question whether that verification has been supplied.

A.1 A separating model for interface completeness

A small model makes the nonvacuity of the quotient theorem explicit. Let $B = \{b_0, b_1\}$ be two original trial records with distinct identified occurrences and let $a(b_0) = 0$, $a(b_1) = 1$ be their fixed original qualification values. Identify each b by its original trial occurrence, independently of its qualification value. With questions q_{id} and q_{qual} , define the original incidences by

$$J(b, q_{\text{id}}, v) \iff v = b, \quad J(b, q_{\text{qual}}, u) \iff u = a(b).$$

Their result types are B and $\{0, 1\}$, respectively. Each question has exactly one result by these independently supplied data. Let K be a nonempty decoration set and $X = B \times K$. The presentation map $(b, k) \mapsto b$ identifies the original bearer without inspecting its qualification. Reading the first coordinate and the supplied trial value gives the two incidence warrants. The complete required queries ask for the original record and its qualification, so

$$P(b, k) = (b, a(b)).$$

Allow a comparison between decorations of the same record: $(b, k)E(b', k')$ exactly when $b = b'$. Then $R = E$, and the maximal complete quotient is canonically B . The group $G = \text{Sym}(K)$ acts by $g(b, k) = (b, gk)$, so every permutation acts lawfully on the decoration coordinate. Since this action is transitive on each decoration fibre, its orbit quotient is also B . No decorative change alters the original qualification, and both admission polarities are realized.

An interface $O(b, k) = b$ satisfies the complete criterion. An interface retaining only $a(b)$ is sufficient in this two-record model because the two values happen to distinguish the records; this is a fact of the model, not a general theorem about admission bits. Add a third distinct failed record b_2 with $a(b_2) = 0$. The same one-bit interface still answers qualification but now

fails the record-identification query. Thus even a successful finite test of a binary interface cannot establish universal coverage without the original use specification.

Taking finite sequences of the original records as histories and retaining their prefixes supplies the history-state construction of Example 11.3. A later b_1 does not remove an earlier b_0 . This combines positive and negative qualification, harmless redescription, a nontrivial observation quotient, and historical retention in an explicit model. It is an illustration of the stated interface semantics, not an empirical claim about a particular battery experiment.

A.2 Logical and computational realizations

Three stronger properties require different proofs. First-order explicit definability requires an appropriate elementary theory and the full implicit-definability premise; the ordinary Beth theorem does not follow from uniqueness on a selected intended subclass of models [12]. Effective numerical extraction requires an algorithm and its correctness or approximation bounds. Physical realization requires an existence and identification theorem in its domain. None is implied merely by quotient factorization, and none is needed to prove same-incidence conservation.

This separation preserves rather than restricts the semantic theorem’s generality. A complete profile can exist without a finite description, and a finite certificate implementation can be justified without claiming that every possible warranted use has such an implementation. No machine-checked formalization of the complete present manuscript is asserted; the proofs given here are the mathematical basis for such a formalization.

B Proof obligations for downstream use

The following ledger records what an application must establish before invoking a boundary conclusion. It is a map of theorem hypotheses, not a procedure that decides them automatically.

Claim	Independent obligation	Exact failure witness
Same-incidence answer	Identify bearer and complete question; establish original incidence and result determinacy	Two warranted unequal results at that locus
Pre-fixational anti-haecceity	Establish unchanged complete content and original-incidence identification; justify equality-only structure where full permutation symmetry is used	A relevant retained relation or target association is omitted
Backward determination	Specify the forgetting map, prove fibre constancy, source coverage, and original answer warrant	One retained source value with different required answers
No circular license	Identify independent grounds and sound rules; show every rule entering the unseeded license family requires a premise in it	An independently grounded entrance, or an incorrect rule-dependency claim
Lawful symmetry	Specify the action from retained data, including all dependent coordinates	An alleged symmetry changes an authoritative datum
Equivariant selection	Give an invariant domain, eligible family, and stabilizer-fixed candidate; expose orbitwise choices	A selected output is moved by an input-fixing symmetry

Claim	Independent obligation	Exact failure witness
Unique equivariant selection	Prove that every eligible stabilizer-fixed set is a singleton	An empty such set or two eligible fixed-point values at one input
Equality-only boundary	Show that equality exhausts authoritative atom structure	A relevant unrepresented relation or anchor
Complete interface	Derive query coverage from intended uses; prove constancy of required answers on interface fibres	Equal interface values requiring different answers
Canonical uniqueness	Prove equality of the relevant quotient kernels and surjectivity onto realized interface values	A lost distinction or retained irreducible excess class
Partial-operation descent	Preserve both original definedness and output equivalence	Equivalent inputs disagree in domain or result
Universal assembly	Fix generators and overlaps independently; establish a colimit in the lawful category	Missing source data or a failed mediator property
Realization uniqueness	Prove candidate coverage, existence, global compatibility, and one lawful assignment orbit	No compatible realization or two inequivalent ones
Historical endpoint test	Prove negativity absorption for the actual transition relation	A legal negative-to-positive edge
Finite compilation	Supply original leaf derivations, application domain and side-condition warrants, and sound local translations	A leaf or constructor application lacks its required derivation or warrant

The ledger exposes a useful asymmetry. Necessary kernel governance is incurred by every actual determinate target. Applying a particular mathematical corollary still requires establishing that target’s corresponding mathematical hypotheses. A metric scale, a partial operator’s domain, and a history-state transition relation are not interchangeable instances of one untyped predicate.

B.1 Two instructive limits on stronger extrapolation

For a finite poset P , a linear extension fixed by every automorphism exists if and only if $\text{Aut}(P)$ is trivial. Indeed, an automorphism preserving a finite linear order fixes every ordered position; conversely, the trivial group fixes every linear extension, and a finite poset has one by repeatedly removing a minimal element. Existence of an invariant extension does not establish uniqueness: the rigid poset $a < b < c$, $a < d$ has three linear extensions. It illustrates the difference between the existence of a fixed point and the existence of exactly one fixed point. The general selector theorem handles both without assuming a total orbit order.

Physical locality likewise does not by itself prohibit additional physical sectors. If outputs at each site depend only on a specified neighborhood, adding an independent sitewise local variable can preserve that locality while changing the theory’s target data. Such a change is not an unchanged presentation of the original complete dynamics, but its difference need not be nonlocal. Excluding it as a physical realization requires a separate completeness theorem for the intended target. The boundary enforces the identity of the claimed object; it does not turn every distinct object into a contradiction.

These examples prevent an exact conservation theorem from being inflated into a universal classification that its premises do not support. They also identify how a stronger domain result could be established: supply the missing uniqueness or physical-exhaustion argument with the same independent warrant and coverage discipline.

C Why the principal mathematical hypotheses cannot be dropped

The following finite and explicit models test the theorem chain at its load-bearing premises. Each preserves the remaining local mathematical structure needed for the stated comparison. They establish necessity of particular qualifications in the mathematical statements, not logical independence of four ontological axioms. The models are themselves determinate constructions and incur the kernel.

C.1 Single-valued incidence and warranted presentation

A determinate bearer does not make every question single-valued. Take a singleton bearer set $B = \{b\}$, two questions q_{id} , q_{out} , and distinct output values 0, 1. Let the identity question return b , while

$$J(b, q_{out}, 0), \quad J(b, q_{out}, 1)$$

both hold. The bearer is identifiable and the possible-output set is determinate. The relation for a single selected output is not functional. Same-incidence answer conservation cannot be applied to that relation. An actual selection or a question asking for the whole possible-output set supplies a different, explicitly identified object of assessment.

Conversely, let $J(b, q, v)$ hold exactly when $v = 0$. A device printing 1 is a determinate device, but its output is not a warranted answer to this original question. Omitting the individual-warrant premise would make the conclusion false. Necessary governance of the device is not a certificate of every claim its output might be used to make. This distinguishes a failed standing-bearing use from the nonexistence of its physical carrier.

Two original questions can also have different answers without any contradiction. If $J(b, q_0, 0)$ and $J(b, q_1, 1)$ are separately warranted, deleting the query index does not establish same-locus discordance. Conversely, inserting an extra index solely to separate two rival answers after they disagree does not establish an independently different locus. The original query and its identification must be supplied before the result comparison.

C.2 Eligibility, partial domains, and genuine symmetry

Let a two-element group fix one input x and act on $Y = \{a, b, c\}$ by exchanging a, b and fixing c . For eligible set $F(x) = \{a, b\}$ there is no equivariant selector, although the ambient output set has the fixed point c . Dropping eligibility would incorrectly turn the irrelevant constant map $s(x) = c$ into an answer to the selection problem. This is why the exact intersection in Theorem 4.6 contains both conditions.

For the domain condition, let the same group exchange two inputs u, v . A rule defined only at u can pass a weak test that compares outputs only when both transformed inputs happen to lie in its domain: for the nonidentity transformation there is no such pair. The test is vacuous. A lawful partial selector must also transport its definedness. Declaring the single chosen input to be its entire domain already makes an unaccounted distinction.

Absence of a named preferred atom does not establish full permutation symmetry. On four atoms arranged in an unmarked cycle, adjacency is an authoritative relation even when no atom is singled out. A permutation carrying an adjacent pair to an opposite pair is not an automorphism of that input. Its adjacency predicate is lawful for the actual automorphism group and is not invariant under every permutation. The general action theorem still applies; the equality-only specialization does not.

C.3 Invariant observations, complete observations, and original comparison

For three positive coordinates, the function x_1/x_2 is invariant under common rescaling. The vectors $(1, 1, 1)$ and $(1, 1, 2)$ have equal values of that function but are not in the same scaling orbit. The full ratio profile in Proposition 10.1 separates them. Thus a coordinate can be perfectly warranted and invariant without constituting a complete invariant for the declared action.

Complete answers likewise need not license every cross-bearer identification. Let $X = \{u, v\}$, let E be equality, and let the only substantive observed answer be $P(u) = P(v) = 0$. The map O with one constant value preserves that answer. It does not preserve the original comparison class. Here $R = E$, so $\ker O \not\subseteq R$. The extra comparison requirement is justified only if the original task requires retaining that distinction; once justified, it cannot be discarded by keeping a success bit.

Finally, preserving all observations is weaker than being a minimal complete interface. Let O retain a complete profile and an additional decorative coordinate. It can be sufficient while its kernel is strictly smaller than R . The quotient theorem does not force its raw representation space to be bijective to the minimal interface. The unique reduction map of Corollary 8.7 records exactly the remaining excess information.

C.4 Definedness and output compatibility are separate

Let $X = \{0, 1\}$ and let $R = X \times X$. Define a partial operation only at 0, with value 0. Whenever two related inputs both lie in its domain, their output classes agree. Nevertheless there is no quotient partial operation reproducing its definedness at every representative: $[0] = [1]$ would have to be both in and outside its domain. Output agreement where defined does not replace the domain condition.

For the converse separation, take $X = \{0, 1, 2\}$ with classes $\{0, 1\}$ and $\{2\}$. Let f be total, with $f(0) = 0$, $f(1) = 2$, and $f(2) = 2$. Domain compatibility holds everywhere, but the equal input classes of 0 and 1 have unequal output classes. Both clauses of Theorem 9.2 therefore do independent work in its exact criterion.

C.5 Retained failure, endpoint standing, and existence

Let the only nontrivial legal edge be $u \rightarrow v$, with $S(u) = 0$ and $S(v) = 1$. The retained path (u, v) is not coherent, although its endpoint is positive. This is the minimal witness that negative absorption is needed for endpoint certification. In the history-state model, the terminal history (u, v) is itself negative under the history predicate, so there is no contradiction between historical retention and a newly positive projected physical stage.

Existence cannot be obtained from uniqueness. The empty eligible family has at most one selector value at each input but has no selector on a nonempty domain. Similarly, an empty result fibre is single-valued but supplies no present result. These cases explain the independent existence clauses in the selection and incidence arguments.

Even compatible-looking components need a joint realization. The sets $\{1, 2\}$, $\{2, 3\}$, and $\{1, 3\}$ have nonempty pairwise intersections and empty total intersection. Giving their elements the same role names does not supply a simultaneous witness. The universal-assembly theorem proves uniqueness for colimits that exist in the specified lawful category; it does not conjure a lawful realization from this pattern of pairwise overlap.

These models supply operationally precise objections. They show which premise has failed and which proposed inference consequently loses its warrant. That specificity strengthens the universal boundary: it prevents both an unwarranted application and an indiscriminate rejection of every construction that differs from a preferred presentation.

References

- [1] Pitts, A. M. (2013). *Nominal Sets: Names and Symmetry in Computer Science*. Cambridge Tracts in Theoretical Computer Science, vol. 57. Cambridge University Press. <https://doi.org/10.1017/CB09781139084673>.
- [2] Gabbay, M. J., and Pitts, A. M. (2002). A new approach to abstract syntax with variable binding. *Formal Aspects of Computing* **13**(3–5), 341–363. <https://doi.org/10.1007/s001650200016>.
- [3] Pitts, A. M. (2003). Nominal logic, a first order theory of names and binding. *Information and Computation* **186**, 165–193. [https://doi.org/10.1016/S0890-5401\(03\)00138-X](https://doi.org/10.1016/S0890-5401(03)00138-X).
- [4] Leinster, T. (2014). *Basic Category Theory*. Cambridge Studies in Advanced Mathematics, vol. 143. Cambridge University Press. <https://doi.org/10.1017/CB09781107360068>. Open author version: <https://arxiv.org/abs/1612.09375>.
- [5] Rutten, J. J. M. M. (2000). Universal coalgebra: a theory of systems. *Theoretical Computer Science* **249**(1), 3–80. [https://doi.org/10.1016/S0304-3975\(00\)00056-6](https://doi.org/10.1016/S0304-3975(00)00056-6).
- [6] Stroud, B. (1968). Transcendental arguments. *The Journal of Philosophy* **65**(9), 241–256. <https://www.jstor.org/stable/2024395>.
- [7] Wiggins, D. (2001). *Sameness and Substance Renewed*. Cambridge University Press. <https://doi.org/10.1017/CB09780511612756>.
- [8] Weatherall, J. O. (2016). Understanding gauge. *Philosophy of Science* **83**(5), 1039–1049. <https://doi.org/10.1086/687936>. Author version with erratum: <https://arxiv.org/abs/1505.02229v2>.
- [9] Ibort, A., Lledó, F., and Pérez-Pardo, J. M. (2015). On self-adjoint extensions and symmetries in quantum mechanics. *Annales Henri Poincaré* **16**, 2367–2397. <https://doi.org/10.1007/s00023-014-0379-4>. Author version: <https://arxiv.org/abs/1402.5537>.
- [10] Ando, T., and Nishio, K. (1970). Positive selfadjoint extensions of positive symmetric operators. *Tohoku Mathematical Journal, Second Series* **22**(1), 65–75. <https://doi.org/10.2748/tmj/1178242861>.
- [11] Kato, T. (1995). *Perturbation Theory for Linear Operators*, 2nd ed. Classics in Mathematics. Springer. <https://doi.org/10.1007/978-3-642-66282-9>.
- [12] Moschovakis, Y. N. (2014). *Lecture Notes in Logic*. Author lecture notes, 29 March 2014. Theorem 3F.3. <https://www.math.ucla.edu/~ynm/lectures/ln1.pdf>.

- [13] Maley, A. J. (2026). *Non-Degenerate Construction and the Kernel of Admissibility*. Reference manuscript, version 49.
- [14] Maley, A. J. (2026). *Constraint Architectures for Coherent Evaluability: Quotient Interfaces, Kernel Instantiation, and Structural Persistence*. Unpublished manuscript.
- [15] Maley, A. J. (2026). *Exact Certificate-Fibre Exhaustion and Unique Role-Occupancy Closure under AASC*. Unpublished manuscript, Module D, 23 August 2026.
- [16] Maley, A. J. (2026). *The Structure of Coherent Construction: A Bivalent Trajectory Theory*. Unpublished manuscript.
- [17] Maley, A. J. (2026). *Metric-Relative Exactness after Non-Primitive Metricity*. Unpublished manuscript, MR1.
- [18] Maley, A. J. (2026). *The Structure of Admissibility: Ametricity, Bivalence, and the Unique Admissible Interior*. Unpublished manuscript, WA version.
- [19] Maley, A. J. (2026). *Metricity as Licensed Description: The L-metric Boundary and the Realized Physical Interior*. Unpublished manuscript.
- [20] Maley, A. J. (2026). *Unus Solus Possibilis Est: Corpus Closure and Uniqueness of the Admissible Interior*. Unpublished manuscript.